A model of geometric K-homology using groupoids

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1 Γ-CW proper complexes

Cita de lo que es un Γ-CW complex

Definition 1.1. *A* Γ*-CW complex is proper when...*

Definition 1.2. *A group is said to acts freely and properly discontinuously on* X *if each element for all* $g \neq e$ *hold that given any* $x \in X$ *there exists an open set* U *in* X *such that* $x \in U$ *and* $(gU) \cap U = \emptyset$.

Remark 1.3. *The equivalence between conditions 1. and 2. in Definition* **??** *is obtained in Theorem 21 in [tD87].*

Revisar si se hace para grupos cualesquera

Definition 1.4. *The following are definitions of a model of a classifying space for proper actions of a group:*

1. *([BCH94], pg 6) A universal example for proper actions of* G*, denoted* $\underline{E}G$*, is a proper* G*-space with the following property: If* X *is any proper* G*-space, then there exists a* G*-map* $\phi_X : X \to \underline{E}G$*, and any two* G*-maps from* X *to* $\underline{E}G$ *are* G*-homotopic.*
2. *([DL17], p. 131) For a family of finite subgroups of* G*, the classifying space for proper actions* $\underline{E}G$ *is a proper* $G - CW$ *complex such that the fixed point set* $\underline{E}G^H$ *is contractible for every* $H \in \mathcal{F}$.
3. *A model of* $\underline{E}\Gamma$ *is a* Γ*-proper space with fixed points relative to subgroups of* Γ *are contractibles.*

Definition 1.5. *A* Γ*-vector bundle* $p : E \to X$ *over a* Γ*-space* X *is a vector bundle* $p : E \to X$ *such that,* E *is a* Γ*-space and* p *is a* Γ*-equivariant map.*

Remark 1.6. *For a* Γ*-vector bundle* $p : E \to X$ *over a proper* Γ*-space* X*, it satisfies that* E *is a* Γ*-proper space: consider a compact set* $K \subset E$*, then* $p(K)$ *is a compact set in* X*, and by the* Γ*-proper structure of* X*, we get that the set*

$$B := \{g \in \Gamma : g\ p(K) \cap p(K) \neq \emptyset\}$$

is finite. Let $A := \{g \in \Gamma : g\ K \cap K \neq \emptyset\}$. We have that $A \subseteq B$, because if $g \in A$, then $\emptyset \neq p(gK \cap K) \subseteq p(gK) \cap p(K) = gp(K) \cap p(K)$, where the last equality is because p is Γ-equivariant.

Definition 1.7. *For X, Y proper Γ-spaces we construct the fiber proper product $X \times_{\underline{E}\Gamma} Y$ as the Γ-equivariant fiber product (or equivariant pullback) of*

$$\phi_X : X \to \underline{E}\Gamma \leftarrow Y : \phi_Y.$$

Lemma 1.8. *([LO01], p. 599) If Γ is a discrete group and $f : X \to Y$ is an equivariant map between finite proper Γ-CW complexes, and $E \to X$ is a Γ-vector bundle. Then There is a Γ-vector bundle $F \to Y$ such that E is summand of f^*F.*

Remark 1.9. *We list two relevant observations about equivariant proper spaces.*

1. *If X is Γ-proper space, and Γ acts trivially on Y, then $X \times Y$ is a Γ-proper space.*
2. *By Remark 1.6. if X is Γ-proper space, $E \to X$ is a vector bundle, then $E \oplus \mathbb{R}^n$ is a proper Γ-vector bundle, where Γ acts trivialy over the real components.*

1.0.1 Groupoids

In this section we discuss about groupoids and groupoid actions where our main reference will be [EM10]. A **groupoid** is a small category such that any arrow is an isomorphism. In this way a groupoid $\mathcal{G}$ is compossed by a class of objects Z and a class of morphisms denoted $\mathcal{G}$ too. Now we present some spaces with groupoid actions.

Definition 1.10. *Let $\mathcal{G}$ be a topological groupoid*[1]*.*

- *A topological space X is called a **$\mathcal{G}$-space** if there is a continuous map $a : X \to Z$ called the **anchor** map and a homeomorphism*

$$\mathcal{G}_s \times_a X \to \mathcal{G}_t \times_a X : (g, x) \mapsto (g, g \cdot x),$$

*which determines the **action**, and holds associativity and unitality conditions.*

- *Definir grupoide propio teneindo en cuenta $\Gamma \ltimes X$*

Example 1.11. *For a group Γ, consider the groupoid $\mathcal{G} := \Gamma^*$, where $\mathcal{G}^0 := \{*\}$ and $\mathcal{G}^1 := \Gamma$.*

Example 1.12. *The transformation groupoid $\mathcal{G} \ltimes X$ (see [EM10], p. 4, Definition 2.2) is defined by X as its object space, morphisms the elements of $\mathcal{G}_s \times_a X$; the range and source maps are defined by $r(g, x) = g \cdot x$ and $s(g, x) = x$ respectively ; and its composition is given by $(g, x) \cdot (h, y) = (gh, y)$.*

[1] A topological groupoid is a groupoid $\mathcal{G}$ such that Z and $\mathcal{G}$ are Hausdorff and the structure maps source and target $s, t : \mathcal{G} \to Z$ are continuous and open (see [FSW20], p. 2).

Definition 1.13. *A $\mathcal{G}$-space X is called* **proper** *if the transformation groupoid $\mathcal{G} \ltimes X$ is a proper groupoid.*

Definition 1.14. *[EM10], p. 9 A properly numerable ver como podemos quitar la palabra numerable $\mathcal{G}$-space $E\mathcal{G}$ is* **universal** *if and only if, any properly numerable $\mathcal{G}$-space X admits a $\mathcal{G}$-map $X \to E\mathcal{G}$ and if there are two $\mathcal{G}$-maps $f, g : X \to E\mathcal{G}$, then f and g are $\mathcal{G}$-equivariantly homotopic.*

With this notion of universal space we have that:

1. If $\mathcal{G}$ is a locally compact groupoid, then there exist a construction of $E\mathcal{G}$.

Remark 1.15. *Let Γ be a group*

1. *If X is a Γ-space, it coincides with the notion of X being a Γ^*-space as a space with a groupoid action.*

2. *X is a Γ-space if and only if X is a $(\Gamma^* \ltimes X)$-space: If X is a Γ-space, define for $x \in X$ and for $(\gamma, y) \in \Gamma \times X$ such that $s(\gamma, y) = x$ (by the fact that the anchor map is the identity of X),*

$$(\gamma, x)x := r((\gamma, x)) = \gamma x;$$

Conversely, if X is a $(\Gamma^* \ltimes X)$-space we define

$$\gamma x := (\gamma, x)x.$$

3. *If X is a Γ-space, and considering the transform groupoid $\Gamma^* \ltimes X$, we get that for any $x \in X$, $\Gamma_x \cong (\Gamma \ltimes X)_x$, this because*

$$\begin{aligned}(\Gamma \ltimes X)_x &= \{(\gamma, x) \in \Gamma^*_s \times_a X : r((\gamma, x)) = x = s((\gamma, x))\} \\ &= \{(\gamma, x) \in \Gamma \times X : \gamma \cdot x = x\} = \Gamma_x\end{aligned}$$

4. *We say that a Γ-space X is proper if the transformation groupoid $\Gamma^* \ltimes X$ is a proper groupoid. When Γ is a discrete group, Parece definicion hay que demostrarlo.*

5. *$\underline{E}\Gamma(\Gamma \ltimes \underline{E}\Gamma) \cong_h \underline{E}\Gamma$ as $\Gamma \ltimes \underline{E}\Gamma$.*

2 Embedding theorems

In order to present and understand the *Factorization Theorem* (see [EM10], p. 18) then, we have to give some preliminaries. First, we remember the following definitions.

Definition 2.1. *Let $\mathcal{G}$ be a topological groupoid.*

- *A* **$\mathcal{G}$-vector bundle** *over a $\mathcal{G}$-space X is a vector bundle with $\mathcal{G}$-action such that the bundle projection, addition and scalar product are $\mathcal{G}$-equivariant.*

- *The* **pull back** *of a $\mathcal{G}$-vector bundle E over Z along the anchor map $a : X \to Z$ to obtain a $\mathcal{G}$-vector bundle over the $\mathcal{G}$-space X is denoted by E^X, and defined by*

$$E^X := X \times_Z E = X_a \times_{\pi_E} E = \{(x, e) \in X \times Z : a(x) = \pi_Z(e)\}$$

A $\mathcal{G}$-vector bundle is called **trivial** *if it is isomorphic to E^X for some $\mathcal{G}$-vector bundle E over Z.*

A $\mathcal{G}$-vector bundle F is called **subtrivial** *if it is a direct summand of a trivial $\mathcal{G}$-vector bundle (i.e., if there exist a $\mathcal{G}$-vector bundle $P \to X$ such that $F \oplus P \cong E^X$).*

Remark 2.2. *When $\mathcal{G}$ is the transformation groupoid (or action groupoid) of a group, we recover the notion of a Γ-vector bundle. For Γ be a discrete group, and X be a Γ-space, there is a bijection:*

$$\{\Gamma\text{−vector bundles over } X\} \leftrightarrow \{(\Gamma^* \ltimes X)\text{−vector bundles over } X\}$$

$(\leftarrow)$: *For a $(\Gamma^* \ltimes X)$-vector bundle $\rho : E \to X$, we note the Γ-vector bundle structure with action on E given by*

$$\gamma e := (\gamma, \rho(e))e,$$

where this action and Remark 1.15 make ρ a Γ-equivariant map

$$\rho(\gamma e) = \rho((\gamma, \rho(e))e) = (\gamma, \rho(e))\rho(e) = \gamma\rho(e).$$

$(\to)$: *If $\rho : E \to X$ is a Γ-vector bundle we note that it has a $(\Gamma^* \ltimes X)$-vector bundle structure by the anchor map $a := \rho : E \to X$, and the natural action*

$$(\gamma, x)e := \gamma e \quad \textit{when } s((\gamma, x)) = a(e), \textit{ i.e., } e \in E_x = \rho^{-1}(x)$$

where this implies that ρ is a $(\Gamma^* \ltimes X)$-equivariant map by

$$\rho((\gamma, x)e) = \rho(\gamma e) = \gamma\rho(e) = \gamma x \quad \Leftrightarrow \quad (\gamma, x)e \in E_{\gamma x} = \rho^{-1}(\gamma x).$$

Definition 2.3 ([EM10], p. 10)**.** *Let $\mathcal{G}$-be a topological groupoid and let X be a numerably proper $\mathcal{G}$-space.*

- *There are* **enough $\mathcal{G}$-vector bundles** *on X if for every $x \in X$ and every finite dimensional representation*[2] *of the stabilizer $\mathcal{G}_x^x$, there is a $\mathcal{G}$-vector bundle over X whose fibre at x contains the given representation of $\mathcal{G}_x^x$.*
- *A $\mathcal{G}$-vector bundle V on X is* **full** *if for every $x \in X$, the fibre V_x contains all irreducible representations of the stabilizer $\mathcal{G}_x^x$.*

Remark 2.4. *We list some important facts that will be useful about the proof of Theorem 2.5 in [EM10].*

[2] An introduction on representations of groups can be consulted in [Ser77].

1. *The vector bundle E^W is the $\mathcal{G}$-vector bundle with total space $Y \times_Z E$, which coincides with the pullback through the maps a and the bundle projection ρ_E:*

$$\begin{tikzcd} E^W := Y \times_Z E \arrow[r] \arrow[d] & E \arrow[d, "\rho_E"] \\ Y \arrow[r, "a"'] & Z \end{tikzcd}$$

This means that if $Z = \{*\}$, then any vector bundle $E \to Z$ must be trivial, $E = \{*\} \times \mathbb{R}^n$, and therefore $E^Y = Y \times_Z E = Y \times (\{*\} \times \mathbb{R}^n) \cong Y \times \mathbb{R}^n$ is also a trivial bundle over Y.

2. *The $\mathcal{G}$-vector bundle $E \to Z$ is picked such that we can get a fibrewise smooth embedding $X \hookrightarrow E$ (this is guaranteed by Theorem 3.22 in [EM10] when X is a smooth $\mathcal{G}$-manifold, such that $\mathcal{G}\backslash X$ has finite covering dimension and there is a full $\mathcal{G}$-vector bundle on Z).*

$$X \hookrightarrow E \qquad \Rightarrow \qquad \begin{tikzcd} & E^Y = a^*E \arrow[d] & E \arrow[d] \\ X \arrow[r, "f"] & Y \arrow[r, "a"] & Z \end{tikzcd}$$

Note that the choice of E is independent of the space Y.

3. *The total space V of the vector bundle $V \to X$ is the normal bundle[3] $\nu_{f\times g}$ of the embedding $(f \times g) : X \to E^Y$ constructed using the embedding $g : X \hookrightarrow E$.*

Now, we present an argument that let us to use the theory of Emmerson - Meyer, mainly Theorem 2.5 in the principal objects of our problem of interest. First we present an important result of Luck.

reparafrasearlo con grupoide $\Gamma \ltimes \underline{E}\Gamma$

Theorem 2.5 ([EM10], p. 18)**.** *Let $\mathcal{G}$ be a proper groupoid, let X and Y be smooth $\mathcal{G}$-manifolds, and let $f : X \to Y$ be a smooth $\mathcal{G}$-equivariant map. Suppose that*

- *Z has enough $\mathcal{G}$-vector bundles and $\mathcal{G}\backslash X$ is compact.*

Then, there are

- *a smooth $\mathcal{G}$-vector bundle V over X,*
- *a smooth $\mathcal{G}$-vector bundle E over Z,*
- *a smoth $\mathcal{G}$-equivariant, open embedding $\eta_f : V \to E^Y$,*

[3] For an introduction to normal bundles, see [GP10].

such that $f = \rho_{E^Y} \circ \eta_f \circ \xi_V$, where $\xi_V : X \to V$ is the zero-section of the fiber bundle $\rho_V : V \to X$.

$$\begin{tikzcd} V \arrow[r, "\eta_f"] \arrow[d, "\rho_V"'] & E^Y \arrow[d, "\rho_{E^Y}"] & E \arrow[d] \\ X \arrow[r, "f"'] & Y \arrow[r, "a"'] & Z \end{tikzcd}$$

3 Useful constructions

3.1 Connected sum generalization

In [Kos92], p. 99, we found a generalization of connected sum of manifolds called the *joining manifolds along submanifolds with boundary* (also called the *pasting*) which is as follows: Consider

1. M_1 and M_2 are two $(n+k)$-dimensional smooth manifolds,
2. $E \to N$ being a k-dimensional Riemannian vector bundle over the n-dimensional closed compact submanifold N,
3. define $\alpha_E : E \setminus \xi_E(N) \to E \setminus \xi_E(N)$ by $\alpha_E(||v||) = \alpha(v) \cdot \frac{v}{||v||}$, where $\alpha : (0,\infty) \to (0,\infty)$ an orientation reversing diffeomorphism,
4. $h_i : E \to M_i$ are Γ-embeddings such that $h_i|_{\xi_E(N)}$ is a neat embedding where $h_i(E)$ are neat tubular neighborhoods of $h_i(\xi_E(N))$.

With these, we define $M(h_1, h_2)$ the join manifold along N as $M_1 \cup M_2 / \sim$, where for $v \in h_1(E \setminus \xi_E(N))$ we consider the identification $v \sim h_2 \circ \alpha_E \circ h_1^{-1}(v)$. We get that $M(h_1, h_2)$ is an oriented smooth manifold with boundary (see [Kos92], p. 100).

Remark 3.1. *Recall that if $\rho : E \to W$ is a Γ-vector bundle, then $TE \cong TW \oplus E$: we have the exact sequence $0 \to \rho^*E \to TE \to \rho^*TW \to 0$ (see [Spi99], p. 103), then the restricted exact sequence $0 \to \rho^*E|_W \to TE|_W \to \rho^*TW|_W \to 0$, where $W \subseteq E$ by the zero section s, is identified with $0 \to E \to TE|_W \to TW \to 0$, by the fact that $\rho^*E|_W \cong E$ and $(\rho^*TW)|_W \cong TW$. We have that this sequence split, because the map $d\rho : TE|_W \to TW$ has right inverse by $\rho \circ s = id_W$.*

3.2 Deformation to the normal cone

Let be M a manifold and $N \subseteq M$ a submanifold of M. The deformation to the normal cone of N in M (see [DS19]) is

$$DNC(M, N) = (M \times \mathbb{R}^*) \sqcup (\nu_N^M \times \{0\})$$

as set and with smooth structure given by the diffeomorphism (onto its image) $\Theta : W' \to M \times \mathbb{R}$, where $W' = \{(x, \xi, t) \in \nu_N^M \times \mathbb{R} : (x, t\xi) \in U'\}$ is an open neighbourhood of $\nu_N^M \times \{0\}$ in $\nu_N^M \times \mathbb{R}$, and where Θ is defined using an exponential map $\theta : U' \subseteq \nu_N^M \to U \subseteq M$ as

$$\Theta(x,\xi,t) = \begin{cases} (\theta(x,t\xi),t) & \text{for } t \neq 0 \\ (x,\xi,0) & \text{for } t = 0 \end{cases}$$

Example 3.2. *Consider a Γ-equivariant smooth map $f : \partial M \to W$ for M and W Γ-manifolds with boundary. By Theorem 2.5, we have the commutative diagram*

$$\begin{tikzcd}
V \arrow[r, "\eta_f"] & E \arrow[d, "\rho_E"] \\
\partial M \arrow[u, "\xi_{\partial M}"] \arrow[ur, dashed, "\tilde{g}"] \arrow[r, "f"'] & W
\end{tikzcd}$$

where η_f is a Γ-embedding. Then in this case

$$DNC(E,\partial M)|_{[0,1]} = (E \times (0,1]) \sqcup (\nu_{\partial M}^E \times \{0\}) = (E \times (0,1]) \sqcup (V \times \{0\}),$$

where $\partial(DNC(E,\partial M)|_{[0,1]}) = (V \times \{0\}) \cup (E \times \{1\}) \cup (E|_{\partial W} \times (0,1))$ Podemos ponerlo un poco mas general.

4 Pushforward homology

Definition 4.1 ([Lüc02])**.** *A Γ-homology theory $\mathscr{H}_*^\Gamma$ with values in R-modules is a collection of covariant functors $\mathscr{H}_n^\Gamma$ from the category of Γ-proper CW cocompact pairs to the category of R-modules, indexed by $n \in \mathbb{Z}$, together with natural transformations called the boundary maps*

$$\partial_n^\Gamma : \mathscr{H}_n^\Gamma(X,A) \to \mathscr{H}_{n-1}^\Gamma(A)$$

for $n \in \mathbb{Z}$ (where $\mathscr{H}_*^\Gamma(A) := \mathscr{H}_*^\Gamma(A,\emptyset)$), such that the following axioms are satisfied:

1. ***Γ-homotopy invariance.***
   If f and g are proper Γ-homotopic maps $(X,A) \to (Y,B)$ of proper Γ-pairs, then the induced maps

   $$f_*, g_* : \mathscr{H}_n^\Gamma(X,A) \to \mathscr{H}_n^\Gamma(Y,B)$$

   are the same for all $n \in \mathbb{Z}$.

2. ***Long exact sequence of a pair.***
   Given a proper Γ-CW cocompact pair (X,A), there is a long exact sequence

   $$\ldots \xrightarrow{j_*} \mathscr{H}_{n+1}^\Gamma(X,A) \xrightarrow{\partial_{n+1}^\Gamma} \mathscr{H}_n^\Gamma(A) \xrightarrow{i_*} \mathscr{H}_n^\Gamma(X) \xrightarrow{j_*} \mathscr{H}_n^\Gamma(X,A) \xrightarrow{\partial_n^\Gamma} \ldots,$$

   where $i : A \to X$ and $j : X \to (X,A)$ are the inclusions.

3. ***Excision.***

   Let (X, A) be a Γ-CW cocompact proper pair and let $f : A \to B$ be a cellular Γ-map of proper Γ-CW cocompact pairs. Equip $(X \cup_f B, B)$ with the induced structure of a proper Γ-CW cocompact pair. Then the canonical map $F : (X, A) \to (X \cup_f B, B)$ induces an isomorphism

$$F_* : \mathscr{H}_n^\Gamma(X, A) \to \mathscr{H}_n^\Gamma(X \cup_f B, B).$$

4. ***Disjoint union axiom.***

   Let $\{X_i \mid i \in I\}$ be a family of proper Γ-CW-complexes. Denote by

$$j_i : X_i \to \coprod_{i \in I} X_i$$

   the canonical inclusion. Then the map

$$\bigoplus_{i \in I} j_i^* : \bigoplus_{i \in I} \mathscr{H}_\Gamma^n(X_i) \to \mathscr{H}_\Gamma^n(\coprod_{i \in I} X_i)$$

   is a group isomorphism.

Definition 4.2 ([Lüc05])**.** *A Γ-cohomology theory $\mathscr{H}_\Gamma^*$ with values in R-modules is a collection of covariant functors $\mathscr{H}_\Gamma^n$ from the category of Γ-pairs to the category of R-modules, indexed by $n \in \mathbb{Z}$, together with natural transformations called the boundary maps*

$$\delta_\Gamma^n : \mathscr{H}_\Gamma^n(A) \to \mathscr{H}_\Gamma^{n+1}(X, A)$$

for $n \in \mathbb{Z}$, such that the following axioms are satisfied:

1. ***Γ-homotopy invariance.***

   If f_0 and f_1 are proper Γ-homotopic maps $(X, A) \to (Y, B)$ of proper Γ-pairs, then the induced maps

$$f_0^n, f_1^n : \mathscr{H}_\Gamma^n(X, A) \to \mathscr{H}_\Gamma^n(Y, B)$$

   are the same for all $n \in \mathbb{Z}$.

2. ***Long exact sequence of a pair.***

   Given a proper Γ-pair (X, A), there is a long exact sequence

$$\ldots \xrightarrow{\delta^{n-1}} \mathscr{H}_\Gamma^n(X, A) \xrightarrow{j^n} \mathscr{H}_\Gamma^n(X) \xrightarrow{i^n} \mathscr{H}_\Gamma^n(A) \xrightarrow{\delta_\Gamma^n} \mathscr{H}_\Gamma^{n+1}(X, A) \xrightarrow{j^{n+1}} \ldots,$$

   where $i : A \to X$ and $j : X \to (X, A)$ are the inclusions.

3. ***Excision.***

   Let (X, A) be a Γ-CW cocompact properpair and let $f : A \to B$ be a cellular Γ-map of proper Γ-pairs. Equip $(X \cup_f B, B)$ with the induced structure of a proper Γ-pair. Then the canonical map $F : (X, A) \to (X \cup_f B, B)$ induces an isomorphism

$$F^n : \mathscr{H}_\Gamma^n(X \cup_f B, B) \to \mathscr{H}_\Gamma^n(X, A).$$

4. ***Disjoint union axiom.***
Let $\{X_i \mid i \in I\}$ be a family of proper Γ-CW-complexes. Denote by

$$j_i : X_i \to \coprod_{i\in I} X_i$$

the canonical inclusion. Then the map

$$\prod_{i\in I} j_i^* : \mathscr{H}_\Gamma^n(\coprod_{i\in I} X_i) \to \prod_{i\in I} \mathscr{H}_\Gamma^n(X_i)$$

is a group isomorphism.

Definition 4.3. *A Γ-cohomology theory $\mathscr{H}_\Gamma^*$ is called multiplicative if holds:*

1. *For any proper Γ-pair (X, A), it satisfy for C_A, a direct set of closed Γ-invariant neighborhoods of A, that*

$$\mathscr{H}_\Gamma^*(X, A) \cong \varinjlim_{U\in C_A} \mathscr{H}_\Gamma^*(X, U).$$

2. *$\mathscr{H}_\Gamma^*$ is endowed with a graded commutative exterior product*

$$\begin{array}{c} \mathscr{H}_\Gamma^i(X_1, A_1) \otimes \mathscr{H}_\Gamma^j(X_2, A_2) \\ \downarrow \\ \mathscr{H}_\Gamma^{i+j}(X_1 \times_{\underline{E}\Gamma} X_2, X_1 \times_{\underline{E}\Gamma} A_2 \cup A_1 \times_{\underline{E}\Gamma} X_2) \end{array}$$

for all $i, j \in \mathbb{Z}$, which are compatible with the boundary maps.

3. *There is a $1 \in \mathscr{H}_\Gamma^0(\underline{E}\Gamma)$ such that for every pair (X, A) and any $\eta \in \mathscr{H}_\Gamma^i(X, A)$, $1 \otimes \eta = \eta \otimes 1 = \eta$.*

Remark 4.4. *The cohomological theories that we will threat are compactly suported cohomology theories, which mean that for not compact spaces, their cohomology will be computed as the relative cohomology of a releated compact space.*

Definition 4.5. *For a d-dimensional real Γ-vector bundle $E \to X$ over a proper Γ-space X, a $\mathscr{H}_\Gamma$-orientation for E is a class $\tau \in \mathscr{H}_\Gamma^d(E)$ such that for each proper Γ-map $f : Y \to X$, multiplication with $f^*\tau \in \mathscr{H}_\Gamma^d(f^*E)$ induces an isomorphism (refered as Thom-isomorphism)*

$$- \otimes \tau : \mathscr{H}_\Gamma^*(Y) \to \mathscr{H}_\Gamma^{*+d}(f^*E).$$

We say that a smooth manifold with boundary M is a $\mathscr{H}_\Gamma$-orientable if TM is $\mathscr{H}_\Gamma$-orientable. When is chosen a fixed $\mathscr{H}_\Gamma$-orientation for a real Γ-vector bundle (or particularly for a manifold), we say that it is a $\mathscr{H}_\Gamma$-oriented bundle (or manifold) Como arregla Luck la orientacion para pareja.

Remark 4.6. *Note that in conditions of Definition 4.5, if we run the multiplicative structure in order to get the map by multiplication of τ, we need that*

$$\mathscr{H}_\Gamma^{*+d}(Y \times_{\underline{E}\Gamma} E) \cong \mathscr{H}_\Gamma^{*+d}(f^*E)$$

Idea:

$$\begin{tikzcd}
Y \times_{\underline{E}\Gamma} E \arrow[d, "\rho_Y"'] & & E \arrow[d, "\rho_E"'] \\
Y \arrow[rr, "f"] \arrow[drr, "a_Y"'] & & X \arrow[d, "a_X"] \\
& & \underline{E}\Gamma
\end{tikzcd}$$

We can consider the model of $\underline{E}\Gamma \cong CX \times \underline{E}\Gamma$ (where the structure of Γ-CW of this space is....) tenemos problemas con punto fijo en CX, utilizar en reemplazo $(X \times [0,1], X \times \{1\})$. With this model we recall that

$$\begin{aligned} Y \times_{\underline{E}\Gamma} E &= \{(y,e) \in Y \times E : a_Y(y) = a_X \circ f(y) = a_X \circ \rho_E(e)\}, \\ f^*E &= \{(y,e) \in Y \times E : f(y) = \rho_E(e)\}. \end{aligned}$$

We denote vector dimension of real smooth vector bundles by $dim_{\mathbb{R}}$, and denote manifold dimension by dim, without the subscript.

Definition 4.7. *A Pushforward structure over a multiplicative Γ-cohomology theory $\mathscr{H}_\Gamma^*$ is an assign for any proper Γ-equivariant, orientation preserving smooth map $f : M \to N$ of manifolds with boundary, a homomorphism*

$$f! : \mathscr{H}_\Gamma^*(M) \to \mathscr{H}_\Gamma^{*+dim(N)-dim(M)}(N),$$

such that

1. *For a composable proper Γ-equivariant, orientation preserving smooth maps we have that $(f \circ g)! = f! \circ g!$ and $Id_X! = Id_{\mathscr{H}_\Gamma^*(X)}$.*

2. *If $F : \mathbb{R} \times M \to N$ is a smooth Γ-equivariant orientation preserving map, then $F(0,-)! = F(1,-)!$.*

3. *If $E \to M$ is a smooth real $\mathscr{H}_\Gamma$-oriented Γ-vector bundle over a manifold with boundary M, then if $s : M \to E$ is the zero section, the pushforward map*

$$s! : \mathscr{H}_\Gamma^*(M) \to \mathscr{H}_\Gamma^{*+dim(E)-dim(M)}(E)$$

   is the Thom isomorphism described in Definition 4.5.

4. *Let N be a manifold with boundary and consider $N^0 = N \setminus \partial N$. If $i^0 : N^0 \to N$ the inclusion, and if U is an open collar around ∂N in N and let $l : \partial N \to U \to N^0$, the inclusion through the collar, then the pushforwards*

$i^0!$ and $l!$ corresponds to the maps appearing in the long exact sequence of the pair $(N, \partial N)$,

$$\cdots \to \mathscr{H}_\Gamma^{*-1}(\partial N) \xrightarrow{l!} \mathscr{H}_\Gamma^*(N^0) \xrightarrow{i^0!} \mathscr{H}_\Gamma^*(N) \to \cdots$$

Here we are using the identification $\mathscr{H}_\Gamma^*(N^0) \cong \mathscr{H}_\Gamma^*(N, \partial N)$. Then, we mean that for the long exact sequence for the maps $i : \partial N \to N$ and $j : N \to (N, \partial N)$, we get that $l! = \delta^{*-1}$ and $i^0! = j^*$.

5. For any $f : (M, \partial M) \to (N, \partial N)$, we had that for the up to homotopy commutative diagram

$$\begin{tikzcd} M \arrow[r, "f"] & N \\ \partial M \arrow[u, "i_{\partial M}"] \arrow[r, "f|_{\partial M}"'] & \partial N \arrow[u, "i_{\partial N}"'] \end{tikzcd}$$

it is satisfied that $i_{\partial N}^* \circ f! = (f|_{\partial M})! \circ i_{\partial M}^*$.

6. Generalization of the Previous Item Let $f : C_1 \to N$ and $g : C_2 \to N$ continuous maps form closet spaces C_1 and C_2. Then for the fiber product

$$\begin{tikzcd} C_1 \arrow[r, "f"] & N \\ C_1 \times_{f,g} C_2 \arrow[u, "\pi_1"] \arrow[r, "\pi_2"'] & C_2 \arrow[u, "g"'] \end{tikzcd}$$

we get that $\pi_2! \circ (\pi_1)^* = g^* \circ f!$.

7. If $f : M_1 \to N_1$ and $g : M_2 \to N_2$, then for the map

$$(f \sqcup g) : M_1 \sqcup M_2 \to N_1 \sqcup N_2,$$

we have that

$$(f \sqcup g)! = f! \oplus g! : \mathscr{H}_\Gamma^*(M_1) \oplus \mathscr{H}_\Gamma^*(M_2) \to \mathscr{H}_\Gamma^*(N_1) \oplus \mathscr{H}_\Gamma^*(N_2)$$

8. If $f_i : M_i \to N$ and $+$ denotes the operation in $\mathscr{H}^*(N)$, then for the map $f_1 \cup f_2 : M_1 \sqcup M_2 \to N$ defined such that $(f_1 \cup f_2) \circ i_k = f_k$, then $(f_1 \cup f_2)! = f_1! + f_2!$.

9. If $i_M : M \to M \sqcup N$ is the natural inclusion, then

$$i_M! : \mathscr{H}_\Gamma^*(M) \to \mathscr{H}_\Gamma^*(M \sqcup N) \cong \mathscr{H}_\Gamma^*(M) \oplus \mathscr{H}_\Gamma^*(N) : x \mapsto x \oplus 0.$$

Definition 4.8. *Let $\mathscr{H}_\Gamma^*$ be a multiplicative Γ-cohomology theory with a pushforward structure. Let (X, A) be a proper Γ-CW pair, let $n \in \mathbb{Z}$, we define the abelian group $\mathscr{H}_*^{pf}(X, A; \Gamma)$ as the set of equivalence relations of tuples $(M, \partial M, \pi, x)$ (called* **cycles***), where:*

1. $(M, \partial M)$ *is a* Γ*-proper manifold possibly with boundary with a* $\mathscr{H}_\Gamma$*-orientation.*

2. $\pi_M : (M, \partial M) \to (X, A)$ *is a continuous* Γ*-equivariant map such that both* $\pi_M|_{M \setminus \pi_M^{-1}(A)} : M \setminus \pi_M^{-1}(A) \to X \setminus A$ *and* $\pi_M|_{\partial M} : \partial M \to A$ *can be factorized as composition* $p \circ f$*, where* p *is a proper map.*

3. $x_M \in \mathscr{H}_\Gamma^n(M)$ *such that* $n = * + dim(M)$*. If* $(M, \partial M)$ *is no a cocompact pair, then* $(M, \partial M, \pi_M, x_M)$ *is a cycle if and only if* $x_M \in \mathscr{H}_\Gamma^n(M) := \mathscr{H}_\Gamma^n(\tilde{M}, \tilde{M} \setminus M)$*, where* $(\tilde{M}, \partial\tilde{M})$ *is a cocompact* Γ*-pair that contains* $(M, \partial M)$ *embedded.*

4. *Two cycles* $(M, \partial M, \pi_M, x_M)$ *and* $(N, \partial N, \pi_N, x_N)$ *are isomorphic cycles if there exist a diffeomorphism* $\phi : M \to N$ *such that* $\pi_M = \pi_N \circ \phi$ *and* $\phi^*(x_N) \cong x_M$*.*

5. *The equivalence relation*

$$(M, \partial M, \pi_M, x_M) \sim (N, \partial N, \pi_N, x_N),$$

is given if there is a k-cycle $(W, \partial W, \pi_W, x_W)$ *and proper smooth,* Γ*-equivariant,* $\mathscr{H}_\Gamma$*-orientation preserving maps* $f_M : (M, \partial M) \to (W, \partial W)$ *and* $f_N : (N, \partial N) \to (W, \partial W)$ *such that,* $\pi_M = \pi_W \circ f_M$*,* $\pi_N = \pi_W \circ f_N$ *and* $f_M!(x_M) = x_W = f_N!(x_N) \in \mathscr{H}_\Gamma^*(W)$*, as shown in the following diagram*

$$\begin{tikzcd}
 & (W, \partial W) \arrow[dd, "\pi_W"] & \\
(M, \partial M) \arrow[ur, "f_M"] \arrow[dr, "\pi_M"'] & & (N, \partial N) \arrow[ul, "f_N"'] \arrow[dl, "\pi_N"] \\
 & (X, A) &
\end{tikzcd}$$

6. *The operation of cycles* $[M, \partial M, \pi, x] + [N, \partial N, \pi', x'] \in \mathscr{H}_*^{pf}(X, A; \Gamma)$ *is defined by*

$$[M \sqcup N, \partial M \sqcup \partial N, \pi \sqcup \pi', x \oplus x'],$$

where $M \sqcup N$ *has the induced* $\mathscr{H}_\Gamma$*-orientation and the cohomology element* $x \oplus y \in \mathscr{H}_\Gamma^{*+dim(M)}(M) \bigoplus \mathscr{H}_\Gamma^{*+dim(N)}(N)$*.*

Definition 4.9. *An homotopy between a pair of maps* $f, g : M \to N$ *is a continuous* Γ*-map* $H : M \times \mathbb{R} \to N$ *such that* $f = H(\cdot, 0)$*,* $g = H(\cdot, 1)$ *and* H *can be factorized by* $p \circ f$ *with* p *being a proper map i.e., for each* $t \in \mathbb{R}$ *the maps* $H_t := H(\cdot, t) : M \to N$ *can be continously factorized as* $p_t \circ f$ *with* p_t *a proper map,*

Remark 4.10. *If* $[M, \partial M, \pi_1, x_M] \in \mathscr{H}_*^{pf}(X, A; \Gamma)$ *and* π_1 *is* Γ*-homotopic to a map* $\pi_2 : M \to X$*, then*

$$[M, \partial M, \pi_1, x_M] = [M, \partial M, \pi_2, x_M].$$

This is by the fact that there is a Γ*-homotopy* $F : M \times \mathbb{R} \to X$ *such that* $F(m, 0) = \pi_1$ *and* $F(m, 1) = \pi_2$ *(with the trivial* Γ*-action on the second factor). Therefore we get the commutative diagram*

$$\begin{tikzcd} & M \times \mathbb{R} \arrow[dd, "F"] & \\ M \arrow[ur, "Id_M \times 0"] \arrow[dr, "\pi_1"'] & & M \arrow[ul, "Id_M \times 1"'] \arrow[dl, "\pi_2"] \\ & X & \end{tikzcd}$$

Since $Id_{M \times \mathbb{R}}$ *is an homotopy between* $Id_M \times 0$ *and* $Id_M \times 1$*, then by the pushforward structure, Definition 4.7, we get that*

$$(Id_M \times 0)!(x_M) = (Id_M \times 1)!(x_M).$$

Remark 4.11. *Considering the following homotopically equivalent* Γ*-CW proper pairs:*

1. (X, A).
2. $(X', A') := (X \times [0, 1], A \times [0, 1])$.
3. $(X'', A'') := (X \times [0, 1], A \times \{0\})$.

Defining $i_X : X \to X'$ *by* $i_X = Id_X \times 1$*, the isomorphism*

$$i_* : \mathscr{H}_*^{pf}(X, A; \Gamma) \to \mathscr{H}_*^{pf}(X', A'; \Gamma)$$

maps a cycle $[M, \partial M, \pi_M, x_M]$ *to* $[M, \partial M, i_X \circ \pi_M, x_M]$*. Now, consider a map* $\varphi : M \to [0, 1]$ *that separate* $\varphi^{-1}(0) = \pi_M^{-1}(A) \supseteq \partial M$ *and* $\varphi^{-1}(1) = M \setminus int(\pi_M^{-1}(A))$[4]*. We define* $\pi'_M : M \to X'$ *by* $\pi_M(m) \times 1$*, such that the following diagram commutes*

$$\begin{tikzcd} M \arrow[d, "\pi_M"'] \arrow[drr, "\pi'_M"] & & \\ (X, A) = (X \times \{1\}, A \times \{1\}) \arrow[rr, "i_X"'] & & (X \times [0, 1], A \times [0, 1]) \end{tikzcd}$$

Therefore, considering the map

$$H' : M \times [0, 1] \to (X', A') : (m, t) \mapsto (\pi_M(m), (1 - t)\varphi(m) + t),$$

[4]This exist because a smooth manifolds is a perfectly normal space, i.e., it is a normal space and every closed set is an intersection of countable open sets.

which if we consider that $\pi_M = p \circ f$ by a proper map $p : R \to X$, then

$$\begin{tikzcd} M \arrow[d, "\pi_M"'] \arrow[dr, "f"] & \\ X & R \arrow[l, "p"] \end{tikzcd} \qquad \Rightarrow \qquad \begin{tikzcd} M \times [0,1] \arrow[d, "H'"'] \arrow[dr, "f\times((1-t)\varphi+t)"] & \\ X \times [0,1] & R \times [0,1] \arrow[l, "p\times Id"] \end{tikzcd}$$

where $p \times Id$ is a proper map, by the fact that proyections are open maps and send compactos to compacts (by continuity), therefore, any covering of a preimage of a compact will be has finite subcover as the union of the finite subcoverings obtained by the proyections of the original cover.

Then, we get an homotopy between $\pi''_M := H'(-,1) = \pi'_M$ and $H'(-,0) : (M, \partial M) \to (X', A')$ which is a pair map, that is, $\pi_M(\partial M) \subseteq A'$ and $[\pi_M(\partial M)]^c \subseteq A'^c$, i.e., $[M, \partial M, \pi''_M, x_M] \in \mathscr{H}^{pf}_*(X'', A''; \Gamma)$ is a cycle such that $\pi''^{-1}_M(A'') \cap M^0 = \emptyset$.

Lemma 4.12. *If $(M, \partial M, \pi_M, x_M) \sim (N, \partial N, \pi_N, x_N)$ by the maps $f_M : M \to W$ and $f_N : N \to W$, then we get that $(M, \partial M, \pi_M, x_M) \sim (N, \partial N, \pi_N, x_N)$ by maps $\phi_M : M \to E^W$ and $\phi_N : N \to E^W$ with ϕ_N being an embedding and $E^W \to W$ is a vector bundle.*

Proof. Applying Theorem 2.5 to the map $f_N : N \to W$, we get that $f_N = \rho_{E^W} \circ \eta_{f_N} \circ \xi_V$ with $\eta_{f_N} : V \to E^W$ is an embedding,

$$\begin{tikzcd} & E^W \arrow[d, "\rho_{E^W}"'] & \\ & W \arrow[dd, "\pi_W"] & V \arrow[ul, "\eta_{f_N}"'] \arrow[d, "\rho_V"] \\ M \arrow[ur, "f_M"] \arrow[dr, "\pi_M"'] & & N \arrow[ul, "f_N"'] \arrow[dl, "\pi_N"] \\ & X & \end{tikzcd}$$

where $\xi_V : N \to V$ denotes the zero section of the vector bundle $\rho : V \to N$. We want to show that $(M, \partial M, \pi_M, x_M) \sim (N, \partial M, \pi_N, x_N)$ through the vector bundle E^W rather than W, using a diagram

$$\begin{tikzcd} & E^W \arrow[dd, "\pi_{E^W}"] & \\ M \arrow[ur, "\phi_M"] \arrow[dr, "\pi_M"'] & & N \arrow[ul, "\phi_N"'] \arrow[dl, "\pi_N"] \\ & X & \end{tikzcd}$$

where, $\phi_M = \xi_{E^W} \circ f_M$, $\phi_N = \eta_{f_N} \circ \xi_V$ and $\pi_{E^W} = \pi_W \circ \rho_{E^W}$. We conclude the proof verifying the following six conditions in order to stablish the relation exhibited in the previous diagram.

- π_{E^W} is a continuous Γ-equivariant map, by composition of continuous Γ-equivariant maps.
- ϕ_M and ϕ_N are proper smooth, Γ-equivariant and $\mathscr{H}_\Gamma$-orientation preserving maps, because zero sections are (locally) smooth such as f_M and η_{f_N} by definition, composition of equivariant maps is equivariant too and $\mathscr{H}_\Gamma$-orientation preserving.
- ϕ_N is an embedding by composition of embeddings, because ξ_V is an embedding by the fact that $\rho_V \circ \xi_V = id_N$ implies that ξ_V is inyective and for any $n \in N$, the derivative $d_n\xi_V$ is inyective by the composition $d_{\xi_V(n)}\rho_V \circ d_n\xi_V = id_{T_nN}$.
- Since $\xi_{E^W} : W \to E^W$ is the zero section, we have that $\rho_{E^W} \circ \xi_{E^W} = id_W$, then $\rho_{E^W}! \circ \xi_{E^W}! = id_W!$ by the pushforward structure, where $\xi_{E^W}!$ is the Thom isomorphism mentioned in Definition 4.7, we get that $\xi_{E^W}! \circ \rho_{E^W}! = id_{E^W}!$, and since $f_N!(x_N) = \rho_{E^W}! \circ \eta_{f_N}! \circ \xi_V!(x_N)$ and $f_M!(x_M) = f_N!(x_N)$, we obtain
$$\begin{aligned} \phi_N!(x_N) &= \eta_{f_N}! \circ \xi_V!(x_N) \\ &= \xi_{E^W}! \circ \rho_{E^W}! \circ \eta_{f_N}! \circ \xi_V!(x_N) \\ &= \xi_{E^W}! \circ f_N!(x_N) \\ &= \xi_{E^W}! \circ f_M!(x_M) \\ &= \phi_M!(x_M). \end{aligned}$$
- We have that $\pi_{E^W} \circ \phi_M = \pi_W \circ (\rho_{E^W} \circ \xi_{E^W}) \circ f_M = \pi_W \circ f_M = \pi_M$.
- We have that $\pi_{E^W} \circ \phi_N = \pi_W \circ (\rho_{E^W} \circ \eta_{f_N} \circ \xi_V) = \pi_W \circ f_N = \pi_N$.

□

Lemma 4.13. *For any Γ-proper embeddings $i_1 : N \to W_1$ and $i_2 : N \to W_2$, there exist Γ-vector bundles $D_1 \to W_1$ and $D_2 \to W_2$ such that*

$$\nu_{\xi_{D_1} \circ i_1} \cong \nu_{\xi_{D_2} \circ i_2}.$$

Proof. Consider the tangent vector bundles $TW_k \to W_k$ for $k = 1, 2$ and the unique up to equivariant homotopy maps $a_1 : W_1 \to Z$, and $a_2 : W_2 \to Z$, by the universal property of $Z = \underline{E}\Gamma$. By the Lemma 1.8, we get that $a_1^*E_1 \cong TW_1 \oplus C_1$ and $a_2^*E_2 \cong TW_2 \oplus C_2$, for Γ-vector bundles $E_1, E_2 \to Z$, $C_1 \to W_1$ and $C_2 \to W_2$, as shows the following diagram:

$$\begin{tikzcd} E_1 \arrow[d] & a_1^*E_1 \cong TW_1 \oplus C_1 \arrow[d] & & a_2^*E_2 \cong TW_2 \oplus C_2 \arrow[d] & E_2 \arrow[d] \\ Z & W_1 \arrow[l, "a_1"'] & N \arrow[l, "i_1"'] \arrow[r, "i_2"'] & W_2 \arrow[r, "a_2"'] & Z \end{tikzcd}$$

We have that Z is the universal proper Γ-CW complex (c.f. [MV03], p. 6) and, by [**?**] we can consider W_i as a compact proper Γ-manifold Falta referencia de esto, making possible the application of Lemma 1.8. If we consider the Γ-vector bundles $D_1 := a_1^* E_2 \oplus C_1$ and $D_2 := a_2^* E_1 \oplus C_2$ over W_1 and W_2, respectively. With this in mind, denoting $E := E_1 \oplus E_2$, we note that:

1. $TD_1|_N \cong TD_2|_N$: by one hand we have $TD_1 \cong TW_1 \oplus D_1$ by Remark 3.1, and by the other hand

$$\begin{aligned}(a_1 \circ i_1)^* E &= i_1^*(a_1^* E_1 \oplus a_1^* E_2) \\ &= i_1^*(TW_1 \oplus C_1 \oplus a_1^* E_2) \\ &\cong i_1^*(TW_1 \oplus D_1) \\ &\cong TW_1|_N \oplus D_1|_N.\end{aligned}$$

   Therefore, $TD_1|_N \cong (a_1 \circ i_1)^* E$, and analogously, $TD_2|_N \cong (a_2 \circ i_2)^* E$. By the universal property of Z, we get that the maps $a_1 \circ i_1, a_2 \circ i_2 : N \to Z$ are Γ-homotopical, then $TD_1|_N \cong TD_2|_N$.

2. $TN \leq TD_i|_N$: Since D_1 is a vector bundle over W_1, using the composition of its zero section and the map i_1, we find that there is an embedding $N \hookrightarrow D_1$, and therefore $TD_1|_N \cong TN \oplus \nu_N^{D_1}$, which implies $TN \leq TD_1|_N$.

Then, we get that $\nu_1 := \nu_N^{D_1} \cong TD_i|_N/TN \cong \nu_N^{D_2} =: \nu_2$ EN caso equivariante si podemos considerar el normal como un cociente?. □

Remark 4.14. *For maps $i_1 : N \to W_1$ and $i_2 : N \to W_2$ being Γ-embeddings, by Lemma 4.13, there exist Γ-vector bundles $D_k \to W_k$ for $k = 1, 2$, such that $\nu_1 := \nu_N^{D_1} \cong TD_i|_N/TN \cong \nu_N^{D_2} =: \nu_2$. With this Γ-equivariant vector bundle isomorphism $\phi : \nu_1 \to \nu_2$, we construct the manifold*

$$W := (D_1 \oplus \mathbb{R})_{\nu_1 \oplus \mathbb{R}} \#_{\nu_2 \oplus \mathbb{R}} (D_2 \oplus \mathbb{R}),$$

the joint of the vector bundles (manifolds) $D_1 \oplus \mathbb{R}$ and $D_2 \oplus \mathbb{R}$ along the submanifold N across the normal bundles $\nu_1 \oplus \mathbb{R}$ and $\nu_2 \oplus \mathbb{R}$, where

$$\partial W = [D_1|_{\partial W_1} \cup D_2|_{\partial W_2}] \oplus \mathbb{R} \setminus [\xi_{\nu_1} \circ i_1(N) \times \{0\}],$$

where ξ_{ν_k} is the zero section of the vector bundle ν_k. This is made by cutting the identified copies of N, $(\xi_{\nu_1 \oplus \mathbb{R}} \circ i_1(N))$ and $(\xi_{\nu_2 \oplus \mathbb{R}} \circ i_2(N))$, where this identification is obtained by the map $i_{\nu_2 \oplus \mathbb{R}} \circ (\phi \times Id_{\mathbb{R}}) \circ (i_{\nu_1 \oplus \mathbb{R}})^{-1}$, using ϕ as a reversing orientation diffeomorphism and $i_{\nu_2 \oplus \mathbb{R}}$ a reversing orientation embedding EN el caso equivariante no se como asegurar su existencia (see [Kos92], pg. 99), making the following diagram commutative being interpreted as a *restricted identity in W, for $r = 1$:*

$$\begin{tikzcd}
D_1 \oplus \mathbb{R} & & D_2 \oplus \mathbb{R} \\
\nu_1 \oplus \mathbb{R} \arrow[u, hook, "i_{\nu_1\oplus\mathbb{R}}"] \arrow[rr, "\phi\oplus Id_{\mathbb{R}}"] & & \nu_2 \oplus \mathbb{R} \arrow[u, hook', "i_{\nu_2\oplus\mathbb{R}}"'] \\
& N \arrow[ul, "(\xi_{\nu_1}\circ i_1)\oplus\{r\}"] \arrow[ur, "(\xi_{\nu_2}\circ i_2)\oplus\{r\}"'] &
\end{tikzcd}$$

Getting $(\xi_{\nu_1} \oplus \{1\}) \circ i_1 = (\xi_{\nu_2} \oplus \{1\}) \circ i_2$ as maps from N to W, and therefore making the commutativity of the following diagram,

$$\begin{tikzcd}
& W \arrow[dd, dashed, "\pi_W"] & \\
W_1 \arrow[ur, "\xi_{D_1}\oplus 1"] & & W_2 \arrow[ul, "\xi_{D_1}\oplus 1"'] \\
M \arrow[u, "i_M"] & N \arrow[ul, "i_1"] \arrow[ur, "i_2"'] & P \arrow[u, "i_P"']
\end{tikzcd}$$

where $\xi_{D_k} \oplus 1|_{i_k(N)} = \xi_{\nu_k} \oplus 1$, for $k = 1, 2$. We claim that

- *W is an oriented smooth manifold (see [Kos92], Chapter VI, Section 4).*
- *W is a Γ-proper space: By Remark 1.6, $D_k \oplus \mathbb{R}$ is a Γ-proper space, and since ϕ is a Γ-equivariant isomorpism, then W is a Γ-proper space by the properness of these bundles.*
- *W is a $\mathscr{H}_\Gamma^*$-oriented space. Pendiente*

Proposition 4.15. *The relation in Definition 4.8 is an equivalence relation.*

Proof. 1. Reflexivity.

$$\begin{tikzcd}
& (M,\partial M) \arrow[dd, "\pi"] & \\
(M,\partial M) \arrow[ur, "Id"] \arrow[dr, "\pi"'] & & (M,\partial M) \arrow[ul, "Id"'] \arrow[dl, "\pi"] \\
& (X,A) &
\end{tikzcd}$$

2. Symmetry: Since the conditions for the relation $\sim$ have no laterally, it is a symmetric relation.

3. Transitivity: Consider the relations $(M, \partial M, \pi_M, x_M) \sim (N, \partial N, \pi_N, x_N)$ given by the maps $i_M : M \to W_1$ and $i_1 : N \to W_1$; and $(N, \partial N, \pi_N, x_N) \sim (P, \partial P, \pi_P, x_P)$ given by $i_P : P \to W_2$ and $i_2 : N \to W_2$. Consider the

space W defined in Remark 4.14 for the maps $i_k : N \to W_k$, getting the following diagram,

$$\begin{tikzcd}
 & W \arrow[dd, dashed, "\pi_W"] & \\
W_1 \arrow[ur, "\xi_{D_1}\oplus 1"] & & W_2 \arrow[ul, "\xi_{D_2}\oplus 1"'] \\
M \arrow[u, "i_M"] & N \arrow[ul, "i_1"'] \arrow[ur, "i_2"] & P \arrow[u, "i_P"']
\end{tikzcd}$$

Since $i_M!(x_M) = i_1!(x_N)$ and $i_2!(x_N) = i_P!(x_P)$, it is satisfied the necessary condition that $((\xi_{D_1}\oplus 1)\circ i_M)!(x_M) = ((\xi_{D_2}\oplus 1)\circ i_P)!(x_P)$, by functoriality of pushforward. We define $\pi_W : W \to N$ by $\pi_W(d \oplus r) = \pi_{W_1}(\rho_1(d))$ if $d \in D_1$, and $\pi_W(d \oplus r) = \pi_{W_2}(\rho_2(d))$ if $d \in D_2$, as in the following diagram:

$$\begin{tikzcd}
D_1 \arrow[d, "\rho_1"'] & D_1 \oplus \mathbb{R} \# D_2 \oplus \mathbb{R} \arrow[ddd, dotted, "\pi_W"] & D_2 \arrow[d, "\rho_2"] \\
W_1 \arrow[ur, "\xi_{D_1}\oplus 1"'] \arrow[ddr, "\pi_{W_1}"'] & & W_2 \arrow[ul, "\xi_{D_2}\oplus 1"] \arrow[ddl, "\pi_{W_2}"] \\
M \arrow[u, "i_M"] \arrow[dr, "\pi_M"'] & N \arrow[ul, "i_1"] \arrow[ur, "i_2"'] & P \arrow[u, "i_P"'] \arrow[dl, "\pi_P"] \\
 & X &
\end{tikzcd}$$

Let us see that π_W is well defined, because if $d = i_{\nu_1}(v)$ for some $v \in \nu_1$, then since $\rho_{D_1} \circ i_{\nu_1} = i_1 \circ \rho_{\nu_1}$

$$\begin{aligned}
\pi_W(d \oplus r) &= \pi_{W_1}(\rho_1(i_{\nu_1}(v))) \\
&= \pi_{W_1} \circ i_1 \circ \rho_{\nu_1}(v) \\
&= \pi_N \rho_{\nu_1}(v)
\end{aligned}$$

following the diagram

$$\begin{tikzcd}
D_1 \arrow[d, "\rho_{D_1}"'] & \nu_1 \arrow[l, "i_{\nu_1}"'] \arrow[rr, "\phi"] \arrow[dr, "\rho_{\nu_1}"] & & \nu_2 \arrow[r, "i_{\nu_2}"] \arrow[dl, "\rho_{\nu_2}"'] & D_2 \arrow[d, "\rho_{D_2}"] \\
W_1 \arrow[drr, "\pi_{W_1}"'] & & N \arrow[ll, "i_1"'] \arrow[rr, "i_2"] \arrow[d, "\pi_N"'] & & W_2 \arrow[dll, "\pi_{W_2}"] \\
 & & X & &
\end{tikzcd}$$

and in the same way, if $d' = i_2(\phi(v)) \in D_2$, then $\pi_W(d' \oplus r) = \pi_N \rho_{\nu_2}(\phi(v)) = \pi_N \rho_{\nu_1}(v)\ \pi_W(d \oplus r)$. Then, we have proved that the cycles $(M, \partial M, \pi_M, x_M)$ and $(P, \partial P, \pi_P, x_P)$ are related by the cycle $(W, \partial W, \pi_W, (\xi_{\nu_1} \circ i_M)!(x_M))$, obtaining the transitivity of the desired equivalence pushforward relation.

$\square$

Remark 4.16. *If there is a proper smooth, Γ-equivariant, $\mathscr{H}_\Gamma$-orientation preserving map $f : (M, \partial M) \to (N, \partial N)$ such that $\pi_N \circ f = \pi_M$ and $f!(x_M) = x_N$, then*

$$[M, \partial M, \pi_M, x_M] = [N, \partial N, \pi_N, x_N],$$

i.e.,

$$[M, \partial M, \pi_N \circ f, x_M] = [N, \partial N, \pi_N, f!(x_M)].$$

The relation is obtained by the following commutative diagram

$$\begin{tikzcd} & N \arrow[dd, dashed, "\pi_N"] & \\ M \arrow[ur, "f"] \arrow[dr, "\pi_M"'] & & N \arrow[ul, "Id"'] \arrow[dl, "\pi_N"] \\ & X & \end{tikzcd}$$

Definition 4.17. *For an $\mathscr{H}_\Gamma$-oriented manifold M with $\mathscr{H}_\Gamma$-orientation $\tau \in \mathscr{H}_\Gamma^{dim(M)}(TM)$, we define $-M$ as the same manifold M but with the $\mathscr{H}_\Gamma$-orientation $-\tau \in \mathscr{H}_\Gamma^{dim(M)}(TM)$.*

Remark 4.18. *Note that $(\mathscr{H}_*^{pf}(X, A; \Gamma), +)$ is actually a group:*

1. *Well defined: If we have*

$$\begin{aligned} [M_1, \partial M_1, \pi_{M_1}, x_{M_1}] &= [M_2, \partial M_2, \pi_{M_2}, x_{M_2}] \\ [N_1, \partial N_1, \pi_{N_1}, x_{N_1}] &= [N_2, \partial N_2, \pi_{N_2}, x_{N_2}], \end{aligned}$$

where the cycles are related by the maps $f_i : M_i \to W_M$ and $g_i : N_i \to W_N$ for $i = 1, 2$, such that $f_1!(x_{M_1}) = f_2!(x_{M_2})$ and $g_1!(x_{N_1}) = g_2!(x_{N_2})$. With this information, we construct the diagram

$$\begin{tikzcd} & W_M \amalg W_N \arrow[dd, dashed, "\pi_{W_M} \amalg \pi_{W_N}"] & \\ M_1 \amalg N_1 \arrow[ur, "f_1 \amalg g_1"] \arrow[dr, "\pi_{M_1} \amalg \pi_{N_1}"'] & & M_2 \amalg N_2 \arrow[ul, "f_2 \amalg g_2"'] \arrow[dl, "\pi_{M_2} \amalg \pi_{N_2}"] \\ & X & \end{tikzcd}$$

By Definition 4.8, in the cohomological elements we have that

$$\begin{aligned} (f_1 \amalg g_1)!(x_{M_1} \oplus x_{N_1}) &= f_1!(x_{M_1}) \oplus g_1!(x_{N_1}) \\ &= f_2!(x_{M_2}) \oplus g_2!(x_{N_2}) \\ &= (f_2 \amalg g_2)!(x_{M_2} \oplus x_{N_2}), \end{aligned}$$

and with this the group operation is well defined.

2. *Neutral element: We have that* $[M, \partial M, \pi_M, 0] \in \mathscr{H}_*^{pf}(X, A; \Gamma)$ *is the zero element, because for* $[N, \partial N, \pi_N, x_N] \in \mathscr{H}_*^{pf}(X, A; \Gamma)$, *we have that*

$$[N, \partial N, \pi_N, x_N] = [N \amalg M, \partial N \amalg \partial M, \pi_N \amalg \pi_M, x_N \oplus 0],$$

by the diagram

$$\begin{tikzcd} & N \amalg M \arrow[dd, dashed, "\pi_N \amalg \pi_M"] & \\ N \arrow[ur, "i_1"] \arrow[dr, "\pi_N"'] & & N \amalg M \arrow[ul, "Id"'] \arrow[dl, "\pi_N \amalg \pi_M"] \\ & X & \end{tikzcd}$$

where $i_1!(x_N) = x_N \oplus 0$.

3. *Inverse element: For a cycle* $[M, \partial M, \pi_M, x_M] \in \mathscr{H}_*^{pf}(X, A; \Gamma)$ *the inverse element is* $[-M, -\partial M, \pi_M, x_M] \in \mathscr{H}_*^{pf}(X, A; \Gamma)$, *because their sum is equal to*

$$[M \amalg -M, \partial M \amalg -\partial M, \pi_M \amalg \pi_{-M}, x_M \oplus x_M].$$

The $\mathscr{H}_\Gamma^*$*-orientation of* M *provides us an* $\mathscr{H}_\Gamma^*$*-orientation of the tangent bundle* $\rho : TM \to M$, *which means that the zero section* $s : M \to TM$ *satisfy that* $s! = - \otimes \tau$, *where the class* $\tau \in \mathscr{H}_\Gamma^m(TM)$ *is such that the map* $- \otimes \tau : \mathscr{H}_\Gamma^*(M) \to \mathscr{H}_\Gamma^{*+m}(TM)$ *is an isomorphism. For* $-M$, *the zero section* $s^- : -M \to T(-M)$ *is the same* s *as a map, but we choose the opposite class satisfying that*

$$s^-! = - \otimes (-\tau) : \mathscr{H}_\Gamma^*(M) \to \mathscr{H}_\Gamma^{*+m}(TM).$$

Then, $s!(x_M) = x_M \otimes \tau = -(x_M \otimes -\tau) = -s^-!(x_M)$. *With this, we define the map* $s \amalg s^- : M \amalg -M \to TM$, *where*

$$(s \amalg s^-)! : \mathscr{H}_\Gamma^*(M \amalg -M) \cong \mathscr{H}_\Gamma^*(M) \oplus \mathscr{H}_\Gamma^*(-M) \to \mathscr{H}_\Gamma^{*+m}(TM)$$

satisfy that $(s \amalg s^-)!(x_M \oplus x_M) = 0 \in \mathscr{H}_\Gamma^*(TM)$. *Using the Remark 4.16 on the equality* $\pi_M \amalg \pi_{-M} = \pi_M \circ \rho \circ (s \amalg s^-)$, *we compute the cycle*

$$\begin{aligned} 0 &= [TM, \partial TM, \pi_M \circ \rho, (s \amalg s^-)!(x_M \oplus x_M)] \\ &= [M \amalg -M, \partial M \amalg -M, \pi_M \circ \rho \circ s \amalg s^-, x_M \oplus x_M] \\ &= [M \amalg -M, \partial M \amalg -M, \pi_M \amalg \pi_{-M}, x_M \oplus x_M]. \end{aligned}$$

Remark 4.19. *If* $[M, \partial M, \pi_M, x], [M, \partial M, \pi_M, y] \in \mathscr{H}_*^{pf}(X, A; \Gamma)$, *with* $x, y \in \mathscr{H}_\Gamma^{*+dim(M)}(M)$ *then*

$$[M, \partial M, \pi_M, x] + [M, \partial M, \pi_M, y] = [M, \partial M, \pi_M, x + y],$$

where $x + y \in \mathscr{H}_\Gamma^{*+dim(M)}(M)$.

Proof. We have to proof the following relation:

$$[M \amalg M, \partial M \amalg \partial M, \pi_M \amalg \pi_M, x \oplus y] = [M, \partial M, \pi_M, x + y]$$

where $x \oplus y \in \mathscr{H}^{*+dim(M)}(M \amalg M) \cong \mathscr{H}^{*+dim(M)}(M) \oplus \mathscr{H}^{*+dim(M)}(M)$. Similarly as in Remark 4.18, we consider $s \amalg s : M \amalg M \to TM$ in order to obtain

$$\begin{aligned}(s \amalg s)!(x \oplus y) &= (s \amalg s)!(x \oplus 0) + (s \amalg s)!(0 \oplus y) = s!(x) + s!(y)\\ &= x \otimes \tau + y \otimes \tau = (x + y) \otimes \tau = s!(x + y),\end{aligned}$$

and with this, in cycles we get that

$$\begin{aligned}[M \amalg M, \partial M \amalg \partial M, \pi_M \amalg \pi_M, x \oplus y] &= [M \amalg M, \partial M \amalg M, \pi_M \circ \rho \circ (s \amalg s), x \oplus y]\\ &= [TM, \partial TM, \pi_M \circ \rho, (s \amalg s)!(x \oplus y)]\\ &= [TM, \partial TM, \pi_M \circ \rho, s!(x + y)]\\ &= [M, \partial M, \pi_M \circ \rho \circ s, x + y]\\ &= [M, \partial M, \pi_M, x + y],\end{aligned}$$

as we desired, where the second and fourth equalities are given by Remark 4.16 applied to $s \amalg s$ and s respectively. $\square$

Definition 4.20. *For a Γ-proper continuous map of pairs $g : (X, A) \to (Y, B)$, define $\mathscr{H}_*^{pf}(g) := g_*$ by*

$$g_* : \mathscr{H}_*^{pf}(X, A; \Gamma) \to \mathscr{H}_*^{pf}(Y, B; \Gamma) : [M, \partial M, \pi, x] \mapsto [M, \partial M, g \circ \pi, x].$$

Proposition 4.21. *The pushforward homology is functorial.*

Proof. We have that $(h \circ g)_* = h_* \circ g_*$ and it is well defined because, if $[M, \partial M, \pi, x] = [N, \partial N, \pi', x'] \in \mathscr{H}_*^{pf}(X, A; \Gamma)$, i.e., there exist a cycle $[W, \partial W, \widehat{\pi}, \widehat{x}]$ over (X, A) and maps $f : (M, \partial M) \to (W, \partial W)$ and $f' : (N, \partial N) \to (W, \partial W)$ such that

$$\begin{aligned}\pi &= \widehat{\pi} \circ f\\ \pi' &= \widehat{\pi} \circ f'\\ f!(x) &= \widehat{x} = f'!(x'),\end{aligned}$$

then composing with g we get

$$\begin{aligned}g \circ \pi &= (g \circ \widehat{\pi}) \circ f\\ g \circ \pi' &= (g \circ \widehat{\pi}) \circ f'\\ f!(x) &= \widehat{x} = f'!(x'),\end{aligned}$$

and we get that $[M, \partial M, g \circ \pi, x] = [N, \partial N, g \circ \pi', x'] \in \mathscr{H}_*^{pf}(Y, B; \Gamma)$, where $g \circ \pi$ and $g \circ \pi'$ are proper Γ-equivariant maps of pairs too by composition, showing that g_* is well defined.

$\square$

Now, we want to prove that the groups $\mathscr{H}_*^{pf}$ define a Γ-equivariant homology theory over the cocompact proper Γ-CW pairs, showing each of the axioms that it has to satisfy (cf. Definition 4.1). This is the main objective of the thesis, and in this work we will spend the rest of the chapter.

Remark 4.22. *When we have an homotopy between CW-spaces maps $X \to Y$, this is regularly identified by a map $[0,1] \times X \to Y$ (cf. [GP10], p.33), but when we deal with homotopies between smooth manifolds $M \to N$, we identify this homotopy with a smooth map $\mathbb{R} \times M \to N$, in order to avoid problems with the boundary and corners.*

Proposition 4.23. *The pushforward homology satisfies Γ-homotopy invariance, i.e., if $g_0, g_1 : (X, A) \to (Y, B)$ are proper Γ-homotopic maps of proper Γ-CW pairs, then for all $n \in \mathbb{Z}$, the induced maps are equal*

$$g_{0*} = g_{1*} : \mathscr{H}_n^{pf}(X, A; \Gamma) \to \mathscr{H}_n^{pf}(Y, B; \Gamma)$$

Proof. Given a Γ-homotopy $H : \mathbb{R} \times (X, A) \to (Y, B)$ between g_0 and g_1, such that $H : X \setminus H^{-1}(B) \to Y \setminus B$ and $H : A \to B$ can be factorizd as $p \circ f$ by a proper p, we have to show that $g_{0*} = g_{1*}$, i.e., for a cycle $[M, \partial M, \pi_M, x_M] \in \mathscr{H}_n^{pf}(X, A; \Gamma)$ we need that

$$[M, \partial M, g_0 \circ \pi_M, x_M] = [M, \partial M, g_1 \circ \pi_M, x_M] \in \mathscr{H}_n^{pf}(Y, B; \Gamma).$$

To prove this, consider the tuple $(M \times \mathbb{R}, \partial M \times \mathbb{R}, \pi_{M\times\mathbb{R}}, x_M)$ and maps $f_0, f_1 : M \to M \times \mathbb{R}$, where

$$\pi_{M\times\mathbb{R}} = H \circ (\pi_M \times Id_{\mathbb{R}}) : (M \times \mathbb{R}, \partial M \times \mathbb{R}) \to (Y, B)$$

and $f_i(m) = (i, m)$ for $i = 0, 1$ and all $m \in M$.

$$\begin{tikzcd}
 & (M \times \mathbb{R}, \partial M \times \mathbb{R}) \arrow[d, "\pi_M \times Id_{\mathbb{R}}"] & \\
(M, \partial M) \arrow[ur, "f_0"] \arrow[dr, "g_0 \circ \pi_M"'] & (X, A) \times \mathbb{R} \arrow[d, "H"] & (M, \partial M) \arrow[ul, "f_1"'] \arrow[dl, "g_1 \circ \pi_M"] \\
 & (Y, B) &
\end{tikzcd}$$

In this case, we get that $\pi_{M\times\mathbb{R}} \circ f_i = g_i \circ \pi_M$ for $i = 0, 1$, and the maps

$$H \circ (\pi_M \times Id)|_{\partial M \times \mathbb{R}} = H|_A \circ (\pi_M \times Id)|_{\partial M \times \mathbb{R}}$$
$$H \circ (\pi_M \times Id)|_{([H\circ(\pi_M \times Id)]^{-1}(B))^c} = H|_{[H^{-1}(B)]^c} \circ (\pi_M \times Id)|_{([H\circ(\pi_M \times Id)]^{-1}(B))^c}$$

hold the condition of being factorized by proper maps. For the cohomology element $x_M \in \mathscr{H}_\Gamma^{n+dim(M)}(M)$, we get that

$$f_0!(x_M) = f_1!(x_M)$$

in $\mathscr{H}_\Gamma^{n+dim(M)+1}(M\times\mathbb{R})$, by considering the proyection $\rho_{M\times\mathbb{R}}: M\times\mathbb{R}\to M$, then

$$(\rho_{M\times\mathbb{R}}\circ f_1)! = (\rho_{M\times\mathbb{R}}\circ f_2)!$$

where by the pushforward structure, Definition 4.7, $\rho_{M\times\mathbb{R}}!$ is an isomorphism and by functoriality $f_1!(x) = f_2!(x)$ Se hace necesario que la Homotopia tambien se trate por factorizacion $p\circ f$ tanto para el cerrado como para el complemento, pues si se pide solo propia no es suficiente para argumentar H restringido a $(H\circ(\pi_M\times Id))^{-1}(B)$ sea propia. $\square$

In order to prove the long exact sequence condition, previously we discuss the following remark.

Remark 4.24. *If we have that $[M,\partial M,\pi_M,x_M] = [N,\partial N,\pi_N,x_N]$ by the cycle $[W,\partial W,\pi_W,x_W]$, then by the Remark 4.16, $[M,\partial M,\pi_M,x_M] = [W,\partial W,\pi_W,x_W]$ in $\mathscr{H}_*^{pf}(X,A;\Gamma)$, with the map $f: M\to W$ satisfying that $\pi_W\circ f = \pi_M$ and $f!(x_M) = x_W$.*

Proposition 4.25. *The pushforward homology satisfies the long exact sequence axiom.*

Proof. For the sequence of cocompact proper Γ-CW pairs

$$(A,\emptyset)\xrightarrow{j}(X,\emptyset)\xrightarrow{i}(X,A)$$

We have to prove that the sequence

$$\mathscr{H}_*^{pf}(A,\emptyset;\Gamma)\xrightarrow{j_*}\mathscr{H}_*^{pf}(X,\emptyset;\Gamma)\xrightarrow{i_*}\mathscr{H}_*^{pf}(X,A;\Gamma)\xrightarrow{\partial}\mathscr{H}_{*-1}^{pf}(A,\emptyset;\Gamma)$$

is an exact sequence, were the maps are defined as

- $j_*([M,\emptyset,\pi_M,x_M]) = [M,\emptyset,j\circ\pi_M,x_M]$.
- $i_*([M,\emptyset,\pi_M,x_M]) = [M,\emptyset,\pi_M,x_M]$.
- $\partial([M,\partial M,\pi_M,x_M]) = [\partial M,\emptyset,\pi_M|_{\partial M},r_\partial(x_M)]$, where $r_\partial = i_{\partial M}^*$ with $i_{\partial M}:\partial M\hookrightarrow M$ the canonical inclusion. This map is well defined, because if $[M,\partial M,\pi_M,x_M] = [N,\partial N,\pi_N,x_N]\in\mathscr{H}_*^{pf}(X,A;\Gamma)$ by the cycle $(W,\partial W,\pi_W,x_W)$ where $f!(x_M) = x_W = g!(x_N)$. From the restriction diagram,

$$\begin{tikzcd} M \arrow[r, "f"] & W & N \arrow[l, "g"'] \\ \partial M \arrow[u, "i_{\partial M}"] \arrow[r, "f|_{\partial M}"'] & \partial W \arrow[u, "i_{\partial W}"'] & \partial N \arrow[u, "i_{\partial N}"'] \arrow[l, "g|_{\partial N}"] \end{tikzcd}$$

by Definition 4.8, we get the commutative diagram

$$\begin{tikzcd} \mathscr{H}_\Gamma^*(M) \arrow[r, "f!"] \arrow[d, "i_{\partial M}^*"'] & \mathscr{H}_\Gamma^*(W) \arrow[d, "i_{\partial W}^*"] & \mathscr{H}_\Gamma^*(N) \arrow[l, "g!"'] \arrow[d, "i_{\partial N}^*"] \\ \mathscr{H}_\Gamma^*(\partial M) \arrow[r, "f|_{\partial M}!"'] & \mathscr{H}_\Gamma^*(\partial W) & \mathscr{H}_\Gamma^*(\partial N) \arrow[l, "g|_{\partial N}!"] \end{tikzcd}$$

Then, we get that $[\partial M, \emptyset, \pi_M|_{\partial M}, r_\partial(x_M)] = [\partial N, \emptyset, \pi_N|_{\partial N}, r_\partial(x_N)]$ in $\mathscr{H}_*^{pf}(A, \emptyset; \Gamma)$, by the cycle $(\partial W, \emptyset, \pi_W|_{\partial W}, i^*_{\partial W}(x_W))$,

$$\begin{aligned} f|_{\partial M}!(i^*_{\partial M}(x_M)) &= i^*_{\partial W}(f!(x_M)) \\ &= i^*_{\partial W}(g!(x_N)) \\ &= g|_{\partial N}!(i^*_{\partial N}(x_N)). \end{aligned}$$

Exactness in $\mathscr{H}_*^{pf}(X, \emptyset; \Gamma)$:

- $Ker(i_*) \subseteq Im(j_*)$: Let $[M, \emptyset, \pi_M, x_M] \in Ker(i_*)$, i.e., it has a related cycle $[N, \partial N, \pi_N, 0] \in \mathscr{H}_*^{pf}(X, A; \Gamma)$ by a Γ-proper manifold pair $(W, \partial W)$, with diagram

$$\begin{tikzcd} & W \arrow[dd, dashed, "\pi_W"] & \\ M \arrow[ur, "f_M"] \arrow[dr, "\pi_M"'] & & N \arrow[ul, "f_N"'] \arrow[dl, "\pi_N"] \\ & X & \end{tikzcd}$$

  such that $f_M!(x_M) = f_N!(0) = 0 \in \mathscr{H}_\Gamma^*(W)$. Since $f_M : (M, \emptyset) \to (W, \partial W)$ can be factorized by

$$M \xrightarrow{f_M^0} W^0 \xrightarrow{i_{W^0}} W$$

  and with this, $f_M^0!(x_M) \in Ker(i_{W^0}!)$, by functoriality of the pushforward. Considering the long exact sequence in the cohomology theory for the pair $(W, \partial W)$, we get

$$\mathscr{H}_\Gamma^{*-1}(\partial W) \xrightarrow{\partial} \mathscr{H}_\Gamma^*(W^0) \xrightarrow{i_{W^0}!} \mathscr{H}_\Gamma^*(W)$$

  By exactness, we have that there exist $y \in \mathscr{H}_\Gamma^{*-1}(\partial W)$ such that $\partial(y) = f_M!(x_M)$. Consider the inclusion through a collar $l : \partial W \to W^0$, and therefore by the definition of pushforward structure, $l! : \mathscr{H}_\Gamma^{*-1}(\partial W) \to \mathscr{H}_\Gamma^*(\partial W^0)$ is identified with the cohomological map ∂. With this, we claim that $j_*([\partial W, \emptyset, \pi_W|_{\partial W}, y]) = [M, \emptyset, \pi_M, x_M]$:

$$\begin{aligned} j_*([\partial W, \emptyset, \pi_W|_{\partial W}, y]) &= [\partial W, \emptyset, j \circ \pi_W|_{\partial W}, y] \\ &= [\partial W, \emptyset, \pi_W \circ l, y] \\ &= [W^0, \emptyset, \pi_W|_{W^0}, l!(y)] \\ &= [W^0, \emptyset, \pi_W|_{W^0}, \partial(y)] \\ &= [W^0, \emptyset, \pi_W|_{W^0}, f_M^0!(x_M)] \\ &= [M, \emptyset, \pi_W \circ f_M, x_M] \\ &= [M, \emptyset, \pi_M, x_M], \end{aligned}$$

where: the first one is the definition of j_*; second is by the fact that $j \circ \pi_W|_{\partial W}$ is homotopic to $j \circ \pi_W \circ l$ by the the homotopy between $i_{\partial W} : \partial W \to W$ and $l : \partial W \to W^0 \subset W$, this homotopy is composed with π_W which is factorized by a proper map, and the compose $j \circ \pi_W = \pi_W$ (and does no change anything in the inverse image of compact sets because $j : A \to X$ is a proper map) ; third one is by Remark 4.16 aplied to the map $l : W^0 \hookrightarrow W$; fourth is by $l! = \partial$; fifth is $\partial(y) = f_M!(x_M)$; sixth is Remark 4.16 applied to f_M^0 and the fact that $\pi_W|_{W^0} \circ f_M^0 = \pi_W \circ f_M$; and the last one is $\pi_M = \pi_W \circ f_M$.

- $Ker(i_*) \supseteq Im(j_*)$: We have to show that for a cycle $[M, \emptyset, \pi_M, x_M] \in \mathscr{H}_*^{pf}(A, \emptyset; \Gamma)$, we have that $i_*(j_*([M, \emptyset, \pi_M, x_M])) = 0$. We have that

$$i_*(j_*([M, \emptyset, \pi_M, x_M])) = [M, \emptyset, j \circ \pi_M, x_M] \in \mathscr{H}_*^{pf}(X, A; \Gamma).$$

We have that this cycle is related with the zero cycle

$$[M \times [0,1], M \times \{1\}, \pi_M \circ \pi_1, 0]$$

by the commutative diagram

$$\begin{tikzcd}
(M, \emptyset) \arrow[rr, "Id \times 0"] \arrow[dr, "j \circ \pi_M"'] & & (M \times [0,1], M \times \{1\}) \arrow[dl, "j \circ \pi_M \circ \pi_1"] \\
& (X, A) &
\end{tikzcd}$$

where we have that $(Id_M \times 0)!(x_M) = 0$ because

$$\mathscr{H}_\Gamma^*(M \times [0,1], M \times \{1\}) = \{0\},$$

by the fact that $i_{M \times \{1\}} : M \times \{1\} \to M \times [0,1]$ is a Γ-homotopy equivalence, therefore in the long exact sequence $i^n_{M \times \{1\}}$ is an isomorphism for all $n \in \mathbb{N}$:

$$\begin{tikzcd}
\cdots \arrow[r] & \mathscr{H}_\Gamma^{*-1}(M \times [0,1]) \arrow[r, "i^{*-1}_{M \times \{1\}}"] & \mathscr{H}_\Gamma^{*-1}(M \times \{1\}) \arrow[dl] \\
& \mathscr{H}_\Gamma^{*}(M \times [0,1], M \times \{1\}) \arrow[dl] & \\
\mathscr{H}_\Gamma^{*}(M \times [0,1]) \arrow[r, "i^{*}_{M \times \{1\}}"] & \mathscr{H}_\Gamma^{*}(M \times \{1\}) \arrow[r] & \cdots
\end{tikzcd}$$

Exactness in $\mathscr{H}_*^{pf}(X, A; \Gamma)$:

- $Ker(\partial) \supseteq Im(i_*)$: For $[M, \emptyset, \pi_M, x_M] \in \mathscr{H}_*^{pf}(X, \emptyset; \Gamma)$, we have that

$$\partial(i_*([M, \emptyset, \pi_M, x_M])) = \partial([M, \emptyset, \pi_M, x_M]) = [\emptyset, \partial\emptyset, \pi_M|_\emptyset, r_\partial(x_M)] = 0,$$

considering that $\mathscr{H}_\Gamma^*(\emptyset) = \{0\}$.

- $Ker(\partial) \subseteq Im(i_*)$: Let $[M, \partial M, \pi_M, x_M] \in Ker(\partial) \subseteq \mathscr{H}_*^{pf}(X, A; \Gamma)$, i.e., that $[\partial M, \emptyset, \pi_M|_{\partial M}, r_\partial(x_M)] = 0 \in \mathscr{H}_*^{pf}(A, \emptyset; \Gamma)$. Then, $[\partial M, \emptyset, \pi_M|_{\partial M}, r_\partial(x_M)] = [W, \emptyset, \pi_W, 0]$ by the existence of a commutative diagram

$$\begin{tikzcd} \partial M \arrow[rr, "f"] \arrow[dr, "\pi_M|_{\partial M}"'] & & W \arrow[dl, "\pi_W"] \\ & A & \end{tikzcd}$$

such that $f!(r_\partial(x_M)) = 0$ and W a manifold without boundary. By Lemma 4.12, we can consider f a Γ-embedding, and then applying Lemma 4.13 to $i_{M^0} \circ l : \partial M \to M$ and $f : \partial M \to W$, to get that there exist Γ-vector bundles $\rho_{D_M} : D_M \to M$ and $\rho_{D_W} : D_W \to W$ such that $\nu_M := \nu_{\xi_{D_M} \circ i_{M^0} \circ l} \cong \nu_{\xi_{D_W} \circ f} =: \nu_W$, and Remark 4.14 in order to define

$$T := (D_M \oplus \mathbb{R})_{\nu_M \oplus \mathbb{R}} \#_{\nu_W \oplus \mathbb{R}} (D_W \oplus \mathbb{R}),$$

with ∂T is the identification of $\partial(D_M) \cong D_M|_{\partial M}$. We have $\xi_{D_M \oplus \mathbb{R}} \circ i_{M^0} \circ l : \partial M \to D_M$ and $\xi_{D_W \oplus \mathbb{R}} \circ f : \partial M \to D_W$ are equals in T, i.e., composing with $i_{D_M \oplus \mathbb{R}}$ and $i_{D_W \oplus \mathbb{R}}$ respectively by Remark 4.14.

With this, defining $\iota : M \to T$ as the natural embedding $i_{D_M \times \{0\}} \circ \xi_{D_M \oplus \mathbb{R}}$, and $\pi_T : T \to X$ is defined by $\pi_T(a) = \pi_M \circ \rho_{D_M \oplus \mathbb{R}}$ if $a \in D_M \oplus \mathbb{R}$, or $\pi_T(a) = \pi_W \circ \rho_{D_W \oplus \mathbb{R}}$ if $a \in D_M \oplus \mathbb{R}$, such as in Proposition 4.15 obtaining in the same way that $\pi_T \circ \iota = \pi_M$. We get that $\iota \circ i_{\partial M} = i_{D_W \times \{0\}} \circ \xi_{D_W \oplus \mathbb{R}} \circ f$, where identifying $i_{\partial T} = i_{D_W \times \{0\}}$, obtaining the commutative diagram

$$\begin{tikzcd} \partial M \arrow[r, "i_{\partial M}"] \arrow[d, "\xi_{D_W \oplus \mathbb{R}} \circ f"'] & M \arrow[d, "\iota"] \\ \partial T \arrow[r, "i_{\partial T}"'] & T \end{tikzcd}$$

therefore by Definition 4.8

$$i^*_{\partial T} \circ \iota!(x_M) = (\xi_{D_W \oplus \mathbb{R}} \circ f)! \circ i^*_{\partial M}(x_M) = 0$$

Since $\iota!(x_M) \in Ker(r_\partial)$, by the long exact sequence in cohomology for the pair $(T, \partial T)$, there exist $y \in \mathscr{H}^*_\Gamma(T^0)$ such that $i_{T^0}!(y) = \iota!(x_M)$, where $i_{T^0} : T^0 \hookrightarrow T$. Then

$$\begin{aligned} i_*([T^0, \emptyset, \pi_T|_{T^0}, y]) &= [T^0, \emptyset, \pi_T|_{T^0}, y] \\ &= [T^0, \emptyset, \pi_T \circ i_{T^0}, y] \\ &= [T, \partial T, \pi_T, i_{T^0}!(y)] \\ &= [T, \partial T, \pi_T, \iota!(x_M)] \\ &= [M, \partial M, \pi_T \circ \iota, x_M] \\ &= [M, \partial M, \pi_M, x_M] \end{aligned}$$

where: first equality is definition of i_*; second is a direct composition; third is By Remark 4.16 on i_{T^0}; fourth is the obtained pushforward equality; fifth is by Remark 4.16 on ι; and sixth is the equality $\pi_M = \pi_T \circ \iota$.

Exactness in $\mathscr{H}_*^{pf}(A, \emptyset; \Gamma)$:

- $Ker(j_{*-1}) \supseteq Im(\partial)$: for $[M, \partial M, \pi_M, x_M] \in \mathscr{H}_*^{pf}(X, A; \Gamma)$, we note that $j_{*-1}(\partial([M, \partial M, \pi_M, x_M])) = [\partial M, \emptyset, j \circ (\pi_M|_{\partial M}), r_\partial(x_M)] \in \mathscr{H}_*^{pf}(X, \emptyset; \Gamma)$ is the zero class, because by the cohomology long exact sequence

$$\mathscr{H}_\Gamma^*(M^0) \xrightarrow{i^0! = j_M^*} \mathscr{H}_\Gamma^*(M) \xrightarrow{i_{\partial M}^*} \mathscr{H}_\Gamma^*(\partial M) \xrightarrow{l! = \partial} \mathscr{H}_\Gamma^{*+1}(M^0)$$

  we have that $l!(r_\partial(x_M)) = 0$, then

$$\begin{aligned}
[\partial M, \emptyset, j \circ (\pi_M|_{\partial M}), r_\partial(x_M)] &= [\partial M, \emptyset, j \circ \pi_M \circ l, r_\partial(x_M)] \\
&= [M^0, \emptyset, j \circ \pi_M, l!(r_\partial(x_M))] \\
&= [M^0, \emptyset, j \circ \pi_M, 0] = 0,
\end{aligned}$$

  where the first equation is by the homotopy given along the collar and the second is Remark 4.16 on $l : \partial M \to M^0$.

- $Ker(j_{*-1}) \subseteq Im(\partial)$: If $[M, \emptyset, \pi_M, x_M] \in ker(j_*) \subseteq \mathscr{H}_*^{pf}(A, \emptyset; \Gamma)$, then $j_*([M, \emptyset, \pi_M, x_M]) = [M, \emptyset, j \circ \pi_M, x_M] = 0$, i.e., by Remark 4.24 there exist a description of the null element $[W, \emptyset, \pi_W, 0_W] \in \mathscr{H}_*^{pf}(X, \emptyset; \Gamma)$ with a map $f : M \to W$ such that $\pi_N \circ f = \pi_M$ and $f!(x_M) = 0_W$.

  Applying the Theorem 2.5 and denoting $V = \nu_M^E$, we get

$$\begin{array}{ccc}
V & \xrightarrow{\eta_f} & E \\
\downarrow & & \downarrow \\
M & \xrightarrow{f} & W
\end{array}$$

  Since $f = \rho_E \circ \eta_f \circ \xi_V$, we have that $\eta_f!(\xi_V!(x_M)) = \xi_E!(f!(x_M)) = 0_E$. Consider

$$T := DNC(E, M)|_{[0,1)} = V \times \{0\} \sqcup E \times (0, 1)$$

  where $\partial T = V \times \{0\}$. Using the map $l : \partial T \to T^0$ homotopic to the tubular neighborhood embedding, since l factors through η_f up to homotopy, we have that $l!(\xi_V!(x_M)) = 0_{T^0}$.

  Consider the cohomology long exact sequence of the pair $(T, \partial T)$,

$$\mathscr{H}_\Gamma^*(T^0) \xrightarrow{i_{T^0}! = j_T^*} \mathscr{H}_\Gamma^*(T) \xrightarrow{r_\partial = i_{\partial T}^*} \mathscr{H}_\Gamma^*(\partial T) \xrightarrow{l! = \partial} \mathscr{H}_\Gamma^{*+1}(T^0)$$

  and since $\mathscr{H}_\Gamma^*(\partial T) \cong \mathscr{H}_\Gamma^*(M)$ by the Thom isomorphism given by the zero section $\xi_V : M \to V$ in the pushforward structure (see Definition 4.7), we get that there exists $y \in \mathscr{H}_\Gamma^*(T)$ such that $r_\partial(y) = \xi_V!(x_M)$. With this we get as we need that $\partial([T, \partial T, \pi_T, y]) = [M, \emptyset, \pi_M, x_M]$, where we define $\pi_T : T \to X$ by

$$\pi_T((x,t)) = \begin{array}{ll} \pi_W \circ \rho_E(x) & \text{if } t \neq 0 \\ \pi_M \circ \rho_V(x) & \text{if } t = 0. \end{array}$$

We show this recalling that $\pi_T|_{\partial T} = \pi_M \circ \rho_V$, and with this

$$\begin{aligned} \partial([T, \partial T, \pi_T, y]) &= [\partial T, \emptyset, \pi_{\partial T}, r_\partial(y)] \\ &= [V, \emptyset, \pi_M \circ \rho_V, \xi_V!(x_M)] \\ &= [M, \emptyset, \pi_M \circ \rho_V \circ \xi_V, x_M] \\ &= [M, \emptyset, \pi_M, x_M], \end{aligned}$$

where the first equality is by definition of ∂; second equality is by $\partial T \cong V$, definition of $\pi_{\partial T} = \pi_T|_{\partial T}$ and $r_\partial(y) = \xi_V!(x_M)$; third equation is by Remark 4.15 applied to ξ_V; and last equation is by ξ_V is a section of $\rho_V : V \to M$.

□

Lemma 4.26. *If $[M, \partial M, \pi_M, x_M] \in \mathscr{H}_*^{pf}(X, A; \Gamma)$ then, there exist a cycle $[W, \partial W, \pi_W, x_W^0] \in \mathscr{H}_*^{pf}(X, A; \Gamma)$ such that $i!(\tilde{x}_M) = x_W^0$ for some $\tilde{x}_M \in \mathscr{H}_\Gamma^*(W, \partial W)$.*

Proof. Consider $l : \partial M \to C^0$ a Γ-embedding given by an open collar $C^0 \subseteq M^0$, and define

$$W := DNC(M^0, C^0) = C^0 \times \{0\} \cup M^0 \times (0,1),$$

which is a smooth manifold with $\partial W = C^0 \times \{0\}$. We can extend the map $l \times \{0\} : \partial \to \partial W$ to a map $\gamma : M \to W$, as follows

1. We extend the map $l : \partial M \to C^0$ to a map $\tilde{l} : M \to M^0$ in such a way that
   (a) $\tilde{l}|_{\partial M} = l$ and $\tilde{l}|_{C^c} = Id_M|_{C^c}$.
   (b) There exist $\widehat{l} : M^0 \to M$ such that $\widehat{l} \circ \tilde{l}$ is homotopic to Id_M.
2. Considering a map $\varphi : M \to [0,1]$ such that $\varphi^{-1}(0) = \partial M$ (which exists by the perfect normality of M), we construct

$$\Gamma : M \to W : m \mapsto \Gamma(m) = (\tilde{l}(m), \varphi(m)).$$

then we get the following diagram

$$\begin{tikzcd} M \arrow[r, "\gamma"] & W & M^0 \times \{1/2\} \arrow[l, "i_{M^0 \times \{1/2\}}"'] \\ \partial M \arrow[u, "i_{\partial M}"] \arrow[r, "l \times \{0\}"'] & C^0 \times \{0\} \arrow[u, "i_{\partial W}"] \arrow[r, "i_{C^0}^{M^0} \times Id_{\{0\}}"'] & M^0 \times \{0\} \arrow[u, "Id_{M^0} \times \{1/2\}"'] \end{tikzcd}$$

where this diagram let us illustrate the following facts:

- $i_{\partial W}: \partial W \to W$ is homotopic to the composition

$$\partial W = C^0 \times \{0\} \xrightarrow{i_{C^0}^{M^0} \times Id_{\{0\}}} M^0 \times \{0\} \xrightarrow{Id_{M^0} \times \{1/2\}} M^0 \times \{1/2\} \xrightarrow{i_{M^0 \times \{1/2\}}} W$$

- Definition 4.7 says that $i_{\partial W}^* \circ \gamma! = (l \times \{0\})! \circ i_{\partial M}^*$ by the left square and

$$[(i_{C^0}^{M^0} \times Id_{\{0\}}) \circ (l \times \{0\})]! \circ i_{\partial M}^* = (i_{M^0 \times \{1/2\}} \circ Id_{M^0} \times \{1/2\})^* \circ \gamma!$$

by the external rectangle. Esta afirmacion depende de que permitamos que el axioma de restricciones no se aplique unicamente a restringir a la frontera, sino a cualesquiera par de restricciones y salvo homotopia.

- We have that $(i_{M^0}^M \circ l)! \circ i_{\partial M}^* = 0$ by the long exact sequence of the pair $(M, \partial M)$. Then, we get the computation

$$\begin{aligned}
i_{\partial W}^* \circ \gamma!(x_M) &= (i_{M^0 \times \{1/2\}} \circ Id_{M^0} \times \{1/2\} \circ i_{C^0}^{M^0} \times Id_{\{0\}})^* \circ \gamma!(x_M) \\
&= (i_{C^0}^{M^0} \times Id_{\{0\}})^* \circ (i_{M^0 \times \{1/2\}} \circ Id_{M^0} \times \{1/2\})^* \circ \gamma!(x_M) \\
&= (i_{C^0}^{M^0} \times Id_{\{0\}})^* \circ [(i_{C^0}^{M^0} \times Id_{\{0\}}) \circ (l \times \{0\})]! \circ i_{\partial M}^*(x_M) \\
&= (i_{C^0}^{M^0} \times Id_{\{0\}})^* \circ 0(x_M) = 0
\end{aligned}$$

Therefore, $\gamma!(x_M) \in Ker(i_{\partial W}^*)$, and by the long exact sequence of the pair $(W, \partial W)$, there exist an element $\tilde{x}_W \in \mathscr{H}(W, \partial W)$ such that $i_{W^0}!(\tilde{x}_W) = \gamma!(x_M)$. Defining $x_M^0 := \gamma!(x_M)$, and $\pi_W : W \to X$ is defined for $(m,t) \in W \subseteq M \times [0,1)$ by

$$\pi_W((m,t)) = \pi_M \circ \widehat{l}(m)$$

by Remark 4.16 applied to $\gamma : M \to W$ we get

$$[M, \partial M, \pi_M, x_M] = [W, \partial W, \pi_W, x_W^0].$$

□

Proposition 4.27. *The pushforward homology satisfy the excision property, i.e., for a proper* Γ*-CW pair* (X, A)*, there exist an isomorphism*

$$\mathscr{H}_*^{pf}(X, A) \cong \mathscr{H}_*^{pf}(X \setminus A).$$

Proof. We have to show that the map

$$\phi_* : \mathscr{H}_*^{pf}(X - A, \emptyset; \Gamma) \to \mathscr{H}_*^{pf}(X, A, \Gamma) : [M, \partial M, \pi_M, x_M] \mapsto [M, \partial M, \phi \circ \pi_M, x_M]$$

is an isomorphism. We define the inverse map

$$h : \mathscr{H}_*^{pf}(X, A, \Gamma) \to \mathscr{H}_*^{pf}(X - A, \emptyset; \Gamma),$$

by $h([M, \partial M, \pi_M, x_M^0]) = [M^0, \emptyset, \pi_M|_{M^0}, \tilde{x}_M]$ where $[M, \partial M, \pi_M, x_M^0]$ is given by Lemma 4.26 such that $i_{M^0}^M!(\tilde{x}_M) = x_M^0$ and such that $\pi_M^{-1}(A) \cap (\partial M)^c = \emptyset$, by Remark 4.11.

- For $[M, \emptyset, \pi_M, x_M] \in \mathscr{H}_*^{pf}(X \setminus A, \emptyset; \Gamma)$ we claim that the computation

$$h \circ \phi_*([M, \emptyset, \pi_M, x_M]) = h([M, \emptyset, \phi \circ \pi_M, x_M]) = [M^0, \emptyset, \phi \circ \pi_M|_{M^0}, x_M^0]$$

coincides with the original cycle, because $M^0 = M$ and $i_{M^0}^M! = Id_{\mathscr{H}_\Gamma^*(M,\emptyset)}$.

- For $[M, \partial M, \pi_M, x_M^0] \in \mathscr{H}_*^{pf}(X, A; \Gamma)$ with $i_{M^0}^M!(\tilde{x}_M) = x_M^0$, therefore

$$\begin{aligned}
\phi_* \circ h([M, \partial M, \pi_M, x_M^0]) &= \phi_*([M^0, \emptyset, \pi_M|_{M^0}, \tilde{x}_M]) \\
&= [M^0, \emptyset, \phi \circ \pi_M \circ i_{M^0}^M, \tilde{x}_M] \\
&= [M^0, \emptyset, \pi_M \circ i_{M^0}^M, \tilde{x}_M] \\
&= [M, \partial M, \pi_M, i_{M^0}^M!(\tilde{x}_M)] \\
&= [M, \partial M, \pi_M, x_M^0]
\end{aligned}$$

where fourth equality is given by Remark 4.16 applied to $i_{M^0}^M : M^0 \to M$.

□

5 Applications

5.1 Pushforward for manifolds without boundary

In this section we are going to define a pushforward structure using the cohomology theory of spaces using the K-theory of C^*-algebras of groupoids. The following are some groupoids that we will have in mind in order to determine the pushforward structures of the cohomology theories that are of our interest.

1. We can consider a smooth manifold M such as a groupoid $M \rightrightarrows_{Id}^{Id} M$ with $s = t = Id_M : M \to M$.

2. If $\rho : E \to M$ is a vector bundle over a smooth manifold $E \to M$, then $E \rightrightarrows M$ with $s = t = \rho$. A particular case that we will use is that if $\rho_E : E \to X$ and $\rho_F : F \to X$ are vector bundles over X, then we will consider the groupoid $E \oplus F \rightrightarrows F$, where $\pi_2 : E \oplus F \to F$ is a vector bundle by the pull back

$$\begin{tikzcd}
\rho_F^* E \cong E \oplus F \arrow[r, "\pi_1"] \arrow[d, "\rho_F^*(\rho_E) \cong \pi_2"'] & E \arrow[d, "\rho_E"] \\
F \arrow[r, "\rho_F"'] & X
\end{tikzcd}$$

where the groupoid operation is given by

$$(v \oplus w)(\tilde{v} \oplus w) = v + \tilde{v} \oplus w, \quad (v \oplus w)^{-1} = -v \oplus w.$$

3. If $f : M \to N$ is a smooth map between smooth manifolds $M^2 \times N \rightrightarrows M \times N$ with $s((x, y), n) = (y, n)$; $t((x, y), n) = (x, n)$;

$$((x, y), n)((y, z), n) = ((x, z), n)$$

4. Applying the normal functor to the groupoid $M^2 \times N \rightrightarrows M \times N$ with the embeddings $(\Delta_M \times f) : M \to M^2 \times N$ in the morpshism level and $(Id_M \times f) : M \to M \times N$, we get the groupoid

$$N(M^2 \times N, M) \rightrightarrows N(M \times N, M)$$

5. Applying the deformation to the normal cone functor to the groupoid $M^2 \times N \rightrightarrows M \times N$ we get

$$DNC(M^2 \times N, M) \rightrightarrows DNC(M \times N, M)$$

6. If $f : M \to N$ is a smooth map, then consider the groupoid

$$TM \ltimes f^*TN \rightrightarrows f^*TN.$$

   This is an example of the groupoid $E \ltimes F \rightrightarrows F$ with E and F being vector bundles over X

$$\begin{tikzcd} E \arrow[r, "\varphi"] \arrow[dr, "\rho_E"'] & F \arrow[d, "\rho_F"] \\ & X \end{tikzcd}$$

   where there is an action of E over F (in fibers) defined by $e \cdot f := \varphi(e) + f \in F$ for $e \in E$ and $f \in F$. Then, for the groupoid $E \ltimes F \rightrightarrows F$, we consider the structure:

   - $\mathcal{G} = E \ltimes F = E \oplus F$ and $\mathcal{G}^0 = F$.
   - $s(e \oplus f) = f$ and $t(e \oplus f) = e \cdot f$.
   - If $s(e_1 \oplus f_1) = t(e_2 \oplus f_2)$, that means $f_1 = e_2 \cdot f$ (and therefore all of the elements are in the fibers of the same $x \in X$), then

$$\mu((e_1 \oplus f_1), (e_2 \oplus f_2)) = (e_1 + e_2) \oplus f_2.$$

   - $i(f) = 0 \oplus f$.
   - $Inv(e \oplus f) = -e \oplus e \cdot f$.

Definition 5.1. *The C^*-algebra of a topological groupoid $\mathcal{G} \rightrightarrows \mathcal{G}^0$, which is denoted by $C^*(\mathcal{G} \rightrightarrows \mathcal{G}^0)$ or $C^*(\mathcal{G})$ when the groupoid structure used is clear, as follows*

1. *Consider the set*

$$C_c^\infty(\mathcal{G}) = \{f : \mathcal{G} \to \mathbb{C} | f \textit{ is smooth and } \overline{supp(f)} \textit{ is compact}\}$$

2. *Consider the maximal norm $||f||_{max} = \sup_\pi ||\pi(f)||_{B(\mathcal{H})}$ for $f \in C_c^\infty(\mathcal{G})$, where π is any representation in a Hilbert space, that is*

$$\pi : C_c^\infty(\mathcal{G}) \to B(\mathcal{H})$$

   and define

$$C^*(\mathcal{G} \rightrightarrows \mathcal{G}^0) := \overline{C_c^\infty(\mathcal{G})}_{||\cdot||_{max}}$$

3. *The product that we will consider is giving by convolution as follows: if $f, g \in C^*(\mathcal{G} \rightrightarrows \mathcal{G}^0)$, then $f * g \in C^*(\mathcal{G} \rightrightarrows \mathcal{G}^0)$ with*

$$(f * g)(\gamma) := \int_{\gamma_1 \circ \gamma_2 = \gamma} f(\gamma_1) g(\gamma_2) d\mu = \int_{\gamma_2 \in \mathcal{G}_{s(\gamma)}} f(\gamma \circ \gamma_2^{-1}) g(\gamma_2) d\mu$$

where μ is a measure Semidensidades y quiza medida de Haar de [Wil19].

Remark 5.2. *We can consider the completion by the reduced norm too, but both completion coincides in the scope of amenability, which will be the back ground of our work. Cita sobre amenabilidad y compleciones*

Now, we are going to present a sequence of arguments in order to construct a morphism $f! : K^*(M) \to K^{*+k}(N)$ in Definition 5.6 when a smooth proper map $f : M \to N$ between smooth manifolds without boundary is given.

First, we can interpret the K-theories of the spaces that apears in the Thom isomorphism of a vector bundle $\rho_E : E \to M$ using groupoids, i.e., the isomorphism

$$Thom : K^*(M) \to K^{*+k}(E)$$

can be described as the groupoid K-theory by

$$Thom : K^*(M \rightrightarrows M) \to K^{*+k}(E \rightrightarrows E)$$

where all source and target maps are identities of the respective space. Then, we get the isomorphism

$$Th : K^*(M \rightrightarrows M) \to K^{*+k}(T^*M \oplus f^*TN \rightrightarrows T^*M \oplus f^*TN)$$

For the groupoid $TM \oplus f^*TN \rightrightarrows f^*TN$ there exist the Fourier transform isomorphism

$$\mathcal{F} : C_0(T^*M \oplus f^*TN \rightrightarrows T^*M \oplus f^*TN) \quad \to \quad C^*(T^*M \oplus f^*TN \rightrightarrows f^*TN)$$

defined for $f \in C_0(T^*M \oplus f^*TN \rightrightarrows T^*M \oplus f^*TN)$ by

$$\mathcal{F}(\tilde{v} \oplus \tilde{w}) = \int_{\phi \in T^*_{\rho_{f^*TN}(\tilde{w})} M} e^{i\phi(\tilde{v})} f(\phi \oplus \tilde{w}) d\mu$$

when $\mathcal{F}(f \cdot g) = \mathcal{F}(f) * \mathcal{F}(g)$, that is, the Fourier transform sends a pointwise multiplication of maps to their convolution, wich is the product that we get in the groupoid C^*-algebra. It is injective by isometry, and surjective by density of the Schwarts space that is a subset of the image of Fourier transform Buena referencia, puede ser el Hollmander para detalles y su uso puede ser Montubert o Connes. Therefore, we have an isomorphism $K(\mathcal{F})$

$$K^{*+k}(T^*M \oplus f^*TN \rightrightarrows T^*M \oplus f^*TN) \quad \to \quad K^{*+k}(T^*M \oplus f^*TN \rightrightarrows f^*TN)$$

Remark 5.3. *Considering the groupoid $TM \ltimes f^*TN \rightrightarrows f^*TN$ we get the embedding $\theta : f^*TN \to TM \ltimes f^*TN$, by the zero section, i.e.,*

$$\theta(x, v) = (x, 0) \oplus (x, v).$$

If we compute the normal bundle, we will get

$$\begin{aligned} N(TM \ltimes f^*TN, f^*TN, \xi_{TM\ltimes f^*TN}) &\rightrightarrows N(f^*TN, f^*TN, Id) \\ &= TM \oplus f^*TN \rightrightarrows f^*TN \end{aligned}$$

By the identification obtained in Remark 5.3, we get that $D_\ltimes$ the deformation to the normal cone applied to the groupoid $TM \ltimes f^*TN \rightrightarrows f^*TN$ and the submanifold embedding $\theta : f^*TN \to TM \ltimes f^*TN$ will be

$$D_\ltimes := DNC(TM \ltimes f^*TN, f^*TN) \cong [TM \oplus f^*TN \times \{0\}] \cup [TM \ltimes f^*TN \times (0, 1]]$$

and using this groupoid we get the maps in K-theory

$$K^*(TM \oplus f^*TN) \xleftarrow[ev_0^\ltimes]{} K^*(D_\ltimes) \xrightarrow[ev_1^\ltimes]{} K^*(TM \ltimes f^*TN)$$

where in this case, both $ev_0^\ltimes$ and $ev_1^\ltimes$ are isomorphisms Una cita para este hecho puede estar bien, aunque no vamos a usar que $ev_1^\ltimes$ sea iso.

Remark 5.4. *Although that $K^*(TM \oplus f^*TN) \cong K^*(TM \ltimes f^*TN)$, but in general there is no a map between the C^*-algebras of convolution of the associated groupoids. In $C^*(TM \ltimes f^*TN \rightrightarrows f^*TN)$ the convolution product is*

$$(f * g)(v \oplus w) = \int_{\tilde{v} \in T_{\rho_{TM}(v)}M} f(v - \tilde{v} \oplus \tilde{v} \cdot w)\; g(\tilde{v} \oplus w) d\mu,$$

but in $C^*(TM \oplus f^*TN \rightrightarrows f^*TN)$ the convolution product results

$$(f * g)(v \oplus w) = \int_{\tilde{v} \in T_{\rho_{TM}(v)}M} f(v - \tilde{v} \oplus w)\; g(\tilde{v} \oplus w) d\mu.$$

where the main incidence of the $w \in f^*TN$ component is to relate all the components in this integration to the same fiber $T_{\rho_{TM}(v)}M = T_{\rho_{f^*TN}(w)}M$, but it is fixed during the integration.

Remark 5.5. *We note that there exist a groupoid isomorphism*

$$N(M^2 \times N, M) \rightrightarrows N(M \times N, M) \quad \cong \quad TM \ltimes f^*TN \rightrightarrows f^*TN$$

where $[sus \oplus Id_{TN} - (df \circ \pi_2)] : N(M^2 \times N, M) \to TM \oplus f^*TN$ defined by

$$[sus \oplus Id_{TN} - (df \circ \pi_2)]([(v, w), z]) = (v - w) \oplus z - d_x f(w)$$

is a vector bundle isomorphism (well defined by the quotient structure of the normal bundle, and an isomorphism by the five lemma) such as the following diagram induce on the fibre of $x \sim ((x,x), f(x)) \in M^2 \times N$

$$\begin{tikzcd}
0 \arrow[r] & T_xM \arrow[r, "\Delta_{TM}\times d_x f"] \arrow[d, "Id_{T_xM}"] & T_xM \oplus T_xM \oplus f^*T_{f(x)}N \arrow[r, "\pi"] \arrow[d, "Id"'] & N_{((x,x),f(x))}(M^2 \times N, M) \arrow[r] \arrow[d, "{[sus\oplus Id_{TN}-(df\circ\pi_2)]}"] & 0 \\
0 \arrow[r] & T_xM \arrow[r] & T_xM \oplus T_xM \oplus f^*T_{f(x)}N \arrow[r, "sus\oplus Id_{TN}-(df\circ\pi_2)"'] & T_xM \oplus f^*T_{f(x)}N \arrow[r] & 0
\end{tikzcd}$$

In the same fashion $N(M\times N, M) \cong f^*TN$ *where* $[Id_{f^*(TN)} - (df\circ\pi_1)]([v,z]) = (z - d_xf(v))$ *define the isomorphism*

$$[Id_{f^*(TN)} - (df \circ \pi_1)] : N(M^2 \times N, M) \to TM \oplus f^*TN$$

This map is well defined because if for $x \sim (x, f(x)) \in M \times N$

$$(v_1, z_1) - (v_2, z_2) \in Im(Id_{T_xM} \oplus d_xf)$$

it means that there exist $v \in T_xM$ *such that* $v_1 - v_2 = v$ *and* $z_1 - z_2 = d_xf(v)$*, which is equivalent to equality of the images*

$$z_1 - d_xf(v_1) = z_2 - d_xf(v_2),$$

and it is an isomorphism by the five lemma.

When we have that $\mathcal{G} \rightrightarrows \mathcal{G}^0$ is a saturated groupoid, that is there are a descomposition

$$\mathcal{G} \rightrightarrows \mathcal{G}^0 = (\mathcal{G}_U \rightrightarrows U) \cup (\mathcal{G}_F \rightrightarrows F)$$

where the first component is open and the second, which is its complement, is close of $\mathcal{G} \rightrightarrows \mathcal{G}^0$, we obtain a exact sequenceCita sobre el uso de propiedad universal de la norma maximal, revisar Dixmier

$$0 \longrightarrow C^*(G_U) \longrightarrow C^*(G) \xrightarrow{ev_0} C^*(G_F) \longrightarrow 0$$

Then, in the deformation to the normal cone, we get the exact sequence

$$\begin{tikzcd}
0 \arrow[r] & C^*(M^2 \times N \times (0,1]) \arrow[d] & \\
& C^*(DNC(M^2 \times N, M)) \arrow[d, "ev_0"] & \\
& C^*(N(M^2 \times N, M)) \arrow[r] & 0
\end{tikzcd}$$

and by the six terms exact sequence in K-theory we get the exact sequence

$$\begin{tikzcd}
K^0(M^2 \times N \times (0,1]) \arrow[d] & K^1(N(M^2 \times N, M)) \arrow[l] \\
K^0(DNC(M^2 \times N, M)) \arrow[d, "ev_0"'] & K^1(DNC(M^2 \times N, M)) \arrow[u, "ev_0"'] \\
K^0(N(M^2 \times N, M)) \arrow[r] & K^1(M^2 \times N \times (0,1]) \arrow[u]
\end{tikzcd}$$

Since $(0,1]$ is a contractible space, then we get the contractibility of the algebra $C^*(M^2 \times N \times (0,1]) \cong C_0^\infty((0,1], C^*(M^2 \times N))$, which means that

$$K^*(M^2 \times N \times (0,1]) = 0.$$

Therefore, by the six terms exact sequence we get the isomorphisms

$$ev_0 : K^{*+k}(DNC(M^2 \times N, M)) \to K^{*+k}(N(M^2 \times N, M)), \quad i = 0, 1.$$

For each $t \in [0,1]$ there exist an evaluation map (where for $t \neq 0$ it is no necessarilly an isomorphism)

$$ev_t : K^{*+k}(DNC(M^2 \times N, M)) \to K^{*+k}(M^2 \times N), \quad i = 0, 1.$$

which is the aplication of the K-theory functor over C^*-algebras to the maps

$$ev_t : C^*(DNC(M^2 \times N, M)) \to C^*(M^2 \times N)$$

where for a map

$$\varphi : N_M^{M^2 \times N} \times \{0\} \sqcup \ M^2 \times N \times (0,1] \to \mathbb{C}$$

in $C^*(DNC(M^2 \times N, M))$ we get $ev_t(\varphi) : M^2 \times N \to \mathbb{C}$ with

$$ev_t(\varphi)((x,y),n) := \varphi((x,y),n,t)$$

Finally, since the pair groupoid $M \times M \rightrightarrows M$ is Morita equivalent to the point groupoid $\{Id_{\{*\}}\} \rightrightarrows \{*\}$, then we get an isomorphism (Necesitamos referencia)

$$Mor : K^{*+k}(M^2 \times N) \to K^{*+k}(N)$$

Definition 5.6. *We define the pushforward $f!: K^*(M) \to K^{*+k}(N)$ for a smooth map $f: M \to N$ between smooth manifolds without boundary as the composition proposed in the following diagram*

$$\begin{array}{c}
K^*(M) = K^*(M \rightrightarrows^{Id}_{Id} M) \\
\Big\downarrow{\scriptstyle Th} \\
K^{*+k}(T^*M \oplus f^*TN \rightrightarrows^{Id}_{Id} T^*M \oplus f^*TN) \\
\Big\downarrow{\scriptstyle \mathcal{F}^{-1}} \\
K^{*+k}(TM \oplus f^*TN \rightrightarrows f^*TN) \\
\Big\downarrow{\scriptstyle (ev_0^{\ltimes})^{-1}} \\
K^{*+k}(DNC(TM \ltimes f^*TN, f^*TN) \rightrightarrows DNC(f^*TN, f^*TN, Id)) \\
\Big\downarrow{\scriptstyle ev_1^{\ltimes}} \\
K^*(TM \ltimes f^*TN \rightrightarrows f^*TN) \\
\Big\downarrow{\scriptstyle ev_0^{-1}} \\
K^{*+k}(DNC(M^2 \times N, M, \Delta \times f) \rightrightarrows DNC(M \times N, M, Id_M \times f)) \\
\Big\downarrow{\scriptstyle ev_1} \\
K^{*+k}(M^2 \times N \rightrightarrows M \times N) \\
\Big\downarrow{\scriptstyle Mor} \\
K^{*+k}(N \rightrightarrows^{Id}_{Id} N) = K^*(N)
\end{array}$$

Remark 5.7. *About the previous definition we can point out the following features:*

- *The K-groups releated are in the mayor way belongs to the context of noncommutative geometry becasue they are the K-theory of the C^*-algebra of a non trivial (space) groupoid, except the two releated to the first Thom isomorphism and the last one which is the codomain of the Morita isomorphism.*
- *Almost all the morphism composed are isomorphism except ev_1 the evaluation of the deformation of the normal cone of the embeding $(\Delta, f): M \to M^2 \times N$.*

Remark 5.8. *Ideas que no he aplicado explicitamente*

- *$K(A \times K(\mathcal{H})) \cong K(A)$.*
- *You have to take care about the part that plays the algebra $C^*((0,1])$ in the argument that ev_0 is an isomorphism. $C_0((0,1]) \cong \{0\}$.*

5.2 Pushforward for manifolds with boundary

We define the pushforward of a map of pairs $f : (M, \partial M) \to (N, \partial N)$, i.e, $f^{-1}(\partial N) = \partial M$, using the description of the pushforward of maps between manifolds without boundary, considering the following releated maps:

- $f_0 := f|_{M^0} : M^0 \to N^0$.
- $f_D : D(M) \to D(N) : [m] \mapsto [f(m)]$, which is a well defined map, because $D(M) = M_+ \cup_{\partial M} M_-$ where $M_\pm = M$ are two disjoint copies of M, and $\partial M_+ = \partial M = \partial M_-$ are pointwise identified in $D(M)$.
- $i_M^D : M \to M_+ \subset D(M)$ is the natural inclusion. Since $(M_+)^c = M_-^0 \subset D(M)$, then M_+ is a closed of the double of M, and therefore i_M^D is a proper map. About this map we want to poit out that

  $$(i_M^D)^* : \mathscr{H}^*(D(M)) \to \mathscr{H}^*(M)$$

  is surjective, by an aplication of a Mayer-Vietoris sequence to the double $D(M) = M_+ \cup_{\partial M} M_-$

  $$\mathscr{H}^*(D(M)) \xrightarrow[(i_{M_+}^D)^* \oplus (i_{M_-}^D)^*]{} \mathscr{H}^*(M_+) \oplus \mathscr{H}^*(M_-) \xrightarrow[(i_{\partial M}^+)^* - (i_{\partial M}^-)^*]{} \mathscr{H}^*(\partial M)$$

  because if $\alpha \in \mathscr{H}^*(M)$, we get that $\alpha \oplus \alpha \in Ker((i_{\partial M}^+)^* - (i_{\partial M}^-)^*)$ which by exactness, there exist $D(\alpha) \in \mathscr{H}^*(D(M))$ such that $(i_{M_+}^D)^*(D(\alpha)) = \alpha$.
- $i_{M^0}^D : M^0 \to M_-^0 \subset D(M)$ the canonical inclusion. We claim that the pushforward $i_{M^0}^D!$ coincides with the map that takes a cohomology element $\alpha \in \mathscr{H}^*(M^0)$ and extended it by zero in M_+ to get an element $\hat{\alpha} \in \mathscr{H}^*(D(M))$. Estamos en revision de esta afirmacion

The definition of the pushforward of $f : M \to N$ is defined following the next diagram of exact sequences

$$\begin{tikzcd}
\mathscr{H}^*(M^0) \arrow[r, "i_{M^0}^D!"] \arrow[d, "f_0!"'] & \mathscr{H}^*(D(M)) \arrow[r, "(i_M^D)^*"] \arrow[d, "f_D!"] & \mathscr{H}^*(M) \arrow[r] \arrow[d, "f!"] & 0 \\
\mathscr{H}^*(N^0) \arrow[r, "i_{N^0}^D!"'] & \mathscr{H}^*(D(N)) \arrow[r, "(i_N^D)^*"'] & \mathscr{H}^*(N) \arrow[r] & 0
\end{tikzcd}$$

The left square commutes by the functoriality of the pushforward of manifolds without boundary, and the right square define (in order to commute) the pushforward $f! : \mathscr{H}^*(M) \to \mathscr{H}^*(N)$ by

$$f!(a) := (i_N^D)^* \circ f_D!(b)$$

for $b \in \mathscr{H}^*(D(M))$ such that $(i_M^D)^*(b) = a$. This map is

- well defined: if $a \in \mathscr{H}^*(M)$, consider $b_1, b_2 \in ((i_M^D)^*)^{-1}(a)$, then $b_1 - b_2 \in Ker((i_M^D)^*) = Im((i_{M^0}^D)!)$. For $c \in \mathscr{H}^*(M^0)$ such that $(i_{M^0}^D)!(c) = b_1 - b_2$, we have that $(i_N^D)^* \circ f_D!(b_1) = (i_N^D)^* \circ f_D!(b_2)$, then
$$f_D!(b_1 - b_2) = f_D! \circ (i_{N^0}^D)!(c)$$
and by the bottom exact sequence releated to N, we get that
$$\begin{aligned}(i_N^D)^* \circ f_D!(b_1) - (i_N^D)^* \circ f_D!(b_2) &= (i_N^D)^* \circ f_D! \circ (i_{M^0}^D)!(c) \\ &= (i_N^D)^* \circ (i_{N^0}^D)! \circ f_0!(c) = 0,\end{aligned}$$
which means that $(i_N^D)^* \circ f_D!(b_1) = (i_N^D)^* \circ f_D!(b_2)$.
- is also a group homomorphism by the fact that the involved morphisms are group homomorphisms.

5.3 Review of Pushforward Axioms in K-theory

5.3.1 Functoriality

5.3.2 Homotopy invariance

5.3.3 Pushforward of embeddings

We are going to show that for $f : M \to N$ there exist an easier description of the pushforward $f!$ using a deformation that involves only commutative algebras either than the noncommutative C^*-algebras related to our regular groupoid deformations. We claim that the following diagram commutes

$$\begin{tikzcd}
K^*(M) \arrow[r, "Th"] \arrow[d, "Th"'] & K^*(T^*M \oplus f^*TN) \arrow[d, "(ev_0^{\ltimes})^{-1} \circ ev_1^{\ltimes} \circ \mathcal{F}"] \\
K^*(f^*TN/TM) \arrow[d, "(ev_0^D)^{-1}"'] & K^*(TM \ltimes f^*TN) \arrow[l, "Mor"'] \arrow[d, "(ev_0)^{-1}"] \\
K^*(D(N, M, f)) \arrow[d, "ev_1^D"'] & K^*(D(M^2 \times N, M, (\Delta \times f))) \arrow[l, "Mor"'] \arrow[d, "ev_1"] \\
K^*(N) & K^*(M^2 \times N) \arrow[l, "Mor"]
\end{tikzcd}$$

To prove this, we get that

1. The middle and bottom diagrams are commutative by a Morita equivalence given by a Hilsum-Skandalis isomorphism of
$$D(N, M, f) \cong_{HS} D(M^2 \times N, M, (\Delta \times f))$$
In order to see this, first consider the groupoid space
$$D(N, M, f) \rightrightarrows D(N, M, f)$$

and the deformation functor applyied to the map $\pi_2 : M \times N \to N$ getting

$$D(\pi_2, Id) : D(M \times N, M, Id \times f) \to D(N, M, f)$$

Then, we consider the pullback groupoid ([Wil19], Example 2.37)

$$\begin{tikzcd}
{}^*D(\pi_2, Id)^* D(N, M, f) \arrow[d, shift right, "\pi_1"'] \arrow[d, shift left, "\pi_2"] & D(N, M, f) \arrow[d, shift right] \arrow[d, shift left] \\
D(M \times N, M, Id \times f) \arrow[r, "D(\pi_2, Id)"'] & D(N, M, f)
\end{tikzcd}$$

where we get the identification of this pullback with the pull back of the spaces groupoids

$${}^*D(\pi_2, Id)^* D(N, M, f) = {}^*\pi^*(f^*TN/TM) \sqcup {}^*\pi_2 \times Id^*(N \times (0, 1])$$

getting the saturated groupoid

$$\begin{tikzcd}
f^*TN \times_\pi f^*TN \arrow[dd, shift right] \arrow[dd, shift left] & & M^2 \times N \times (0, 1] \arrow[dd, shift right] \arrow[dd, shift left] \\
& \sqcup & \\
f^*TN & & M \times N \times (0, 1]
\end{tikzcd}$$

We claim that this obtained groupoid is morita equivalent to the deformation groupoid

$$D(M^2 \times N, M, \Delta \times f) \rightrightarrows D(M \times N, M, f)$$

This will be guarantized if we define a generalized isomorphism in zero parts of both groupoids. This is defined by the following generalized morphisms compose

$$\begin{tikzcd}
TM \ltimes f^*TN \arrow[d, shift right] \arrow[d, shift left] & f^*TN \arrow[dl, "Id"] \arrow[dr, "\pi"'] & f^*TN/TM \arrow[d, shift right] \arrow[d, shift left] & f^*TN \arrow[dl, "\pi"] \arrow[dr, "Id"'] & f^*TN \times_\pi f^*TN \arrow[d, shift right] \arrow[d, shift left] \\
f^*TN & & f^*TN/TM & & f^*TN
\end{tikzcd}$$

where the leteral actions are given in the following way

- $TM \ltimes f^*TN f^*TN \curvearrowright f^*TN$ with anchor the identity and for $w \in f^*TN$, then $(v \oplus w) \cdot w = v \ltimes w \in f^*TN$.
- f^*TN/TM has a bilateral trivial action over f^*TN, where the only posible product is for any $w \in f^*TN$,

$$[w] \cdot w = w \cdot [w] := w.$$

- $f^*TN \curvearrowleft f^*TN \times_\pi f^*TN$ with anchor the identity and for $w \in f^*TN$, then $w \cdot (w \oplus \tilde{w}) = \tilde{w} \in f^*TN$.

With these actions we have that both generalized morphisms are invertible by

(a) All anchors are invariant respect the domain groupoid action in each part of the diagram.

(b) All anchors are principal groupoid bundles of the respective domain groupoid action. Poner las cuentas

2. Since the top diagram is the result to apply the K-functor to the respective C^*-algebras to the diagram

$$\begin{tikzcd} M \arrow[r, "s"] \arrow[d, "s"'] \arrow[dr] & T^*M \oplus f^*TN \arrow[r, "\mathcal{F}"] & TM \oplus f^*TN \arrow[d, "in_1"] \\ f^*TN/TM & TM \ltimes f^*TN \arrow[l, dashed, "HS"] & D(TM \ltimes f^*TN, TM \oplus f^*TN) \arrow[l, "ev_0"] \end{tikzcd}$$

where $K^*(s)$ are the inverse of the respective Thom isomorphism, we claim that the both shapes, the right trapezium commutes as inclusions of the groupoid $M \rightrightarrows M$ in each morphism, and the triangle commutes because the Hilsum-Skandalis morphism commutes (as an identity) in the inclusion groupoid of $M \rightrightarrows M$ by the zero sections. Tenemos las cuentas, incluirlas.

5.3.4 Extending by zero interpretation

If we consider in the previous section an embedding given by the inclusion of an open set $U \subseteq M^0$, that is $f = i_U^{M^0} : U \to M^0$ we simplify the previous result, by the fact that $N(M^0, U, i_U^{M^0}) \cong U$, therefore the Tom isomorphism will be replaced by an identity map. In the deformation of the map $i_U^{M^0}$ we get

$$D := DNC(M^0, U, i_U^{M^0}) = U \times \{0\} \sqcup M^0 \times (0,1] = U \times [0,1] \cup M^0 \times (0,1]$$

where the last description is an (not disjoint) union of open subsets, and aplying the Mayer-Vietoris sequence

$$\begin{tikzcd} K^{*-1}(U \times (0,1]) \arrow[d] & & K^*(U \times (0,1]) \\ K^*(D) \arrow[r] & K^*(U \times [0,1]) \oplus K^*(M^0 \times (0,1]) \arrow[ur] & \end{tikzcd}$$

since $K^{*-1}(U \times (0,1]) = K^*(U \times (0,1]) = K^*(M^0 \times (0,1]) = 0$, then we get the restriction isomorphism $K(r) : K^*(D) \to K^*(U \times [0,1]) \cong K^*(U)$. Then we get the commutative diagram

$$\begin{tikzcd} K^*(D) \arrow[r, tail, two heads, "K(r)"] \arrow[dr, "ev_1"'] & K^*(U) \arrow[d, "K(ext)"] \\ & K^*(M^0) \end{tikzcd}$$

where $ext := ev_1 \circ K(r)^{-1}$ is the K-functor applied to the completion of the map that send an $f \in C_c^\infty(U) \subset C^*(M^0)$ to $\hat{f} \in C_c^\infty(D)$ such that $\hat{f}((x,t)) = f(x)$ if $x \in U$ and $\hat{f}((x,t)) = 0$ if $x \in M^0 \setminus U$. Then, composing with ev_1 we get $ev_1(\hat{f})(x) = f(x)$ if $x \in U$ and $ev_1(\hat{f})((x)) = 0$ if $x \in M^0 \setminus U$, i.e., $ext : C_c^\infty(U) \to C_c^\infty(M^0)$ is the extension by zero to all M^0. Falta buen argumento en complesiones sin decir que r es invertible en C^*-algebras, pues esta es solo invertible en K-teoria. Since $r \sim_h ev_0$, we have that $K(r) = K(ev_0)$. Then we have in the evaluation maps

$$K^*(U \times [0,1]) \xleftarrow{ev_0} K^*(D) \xrightarrow{ev_1} K^*(M^0)$$

that $K(ev_0)^{-1} \circ ev_1 = K(r)^{-1} \circ ev_1 = ext$.

Now, if we consider the open inclusion $i_U^M : U \to M$, for $U \subseteq M^0$ an open set, we get that $i_U^M!$ is also an extension by zero, by the construction of pushforward to manigolds with boundary

$$\begin{tikzcd} \mathscr{H}^*(U) \arrow[r, "i_U^D!"] \arrow[d, "(i_U^M)_0!"'] & \mathscr{H}^*(D(U)) \arrow[r, "(i_U^D)^*"] \arrow[d, "(i_U^M)_D!"] & \mathscr{H}^*(U) \arrow[r] \arrow[d, "(i_U^M)!"] & 0 \\ \mathscr{H}^*(M^0) \arrow[r, "i_{M^0}^D!"'] & \mathscr{H}^*(D(M)) \arrow[r, "(i_M^D)^*"'] & \mathscr{H}^*(M) \arrow[r] & 0 \end{tikzcd}$$

we get that the top exact sequence split by the pushforward of the map $i_U^D : U \to D(U) = U \sqcup U$, that is $(i_U^D)^* \circ i_U^D! = Id_{\mathscr{H}^*(U)}$, and with this, we get

$$(i_U^M)! = (i_M^D)^* \circ (i_U^M)_D! \circ (i_U^D)_0!$$

where $(i_U^M)_D! = i_{D(U)}^{D(M)}!$ and both $(i_U^D)_0!$ and $i_{D(U)}^{D(M)}!$ are extensions by zero, then $i_U^M!$ is an extension by zero to M.

5.3.5 Thom isomorphism axiom

In order to verify that for the zero section $s : M \to E$ of a vector bundle $E \to M$, that $s! = \tau_E$,

5.3.6 Long exact sequence

For a pair $(N, \partial N)$ we have the long exact sequence of Definition 4.2

$$\mathscr{H}^*(\partial N) \xrightarrow{\partial} \mathscr{H}^*(N, \partial N) \xrightarrow{j^*} \mathscr{H}^*(N)$$

with $j : (N, \emptyset) \to (N, \partial N)$ and our purpose is to show that $j^* = i_{N^0}!$ No es un pull back, sino un entendimiento de una extension por cero desde N^0 and $\partial = l!$, where $l : \partial N \to C^0 \subset N^0$ where C its a global collar with $C \cong \partial N \times [0, 1)$. Then, we have the following diagram at level of C^*-algebras

$$\begin{tikzcd}
0 \arrow[r] & C^*(\partial N \times (0,1)) \arrow[r] & C^*(\partial N \times [0,1)) \arrow[r] & C^*(\partial N) \arrow[r] & 0 \\
0 \arrow[r] & C^*(DNC(N^0, \partial N, l)) \arrow[r] \arrow[u, "ev_0"] \arrow[d, "ev_1"'] & C^*(\overline{D}) \arrow[r] \arrow[u, "\overline{ev_0}"'] \arrow[d, "\overline{ev_1}"] & C^*(\partial N \times [0,1]) \arrow[r] \arrow[u, "ev_0^h"'] \arrow[d, "ev_1^h"] & 0 \\
0 \arrow[r] & C^*(N^0) \arrow[r] & C^*(N) \arrow[r] & C^*(\partial N) \arrow[r] & 0
\end{tikzcd}$$

where $\overline{D} := (\partial N \times [0, 1)) \times \{0\} \sqcup N \times (0, 1]$. This space is obtained as a restriction of a deformation in the following way:

- Consider a smooth map $z : D(N) = N^+ \cup_{\partial N} N^- \to \mathbb{R}$ such that $z^{-1}(0) = \partial N$ and $z^{-1}([0, \infty)) = N^+$, then we can consider it as a pair map

$$z : (D(N), \partial N) \to (\mathbb{R}, \{0\})$$

- Consider the deformation of the embedding $i_{\partial N} : \partial N \to D(N)$ which is an extension of the map $i_{\partial N} : \partial N \to N$, where $N(D(N), \partial N) = N(N, \partial N) \cong \partial N \times \mathbb{R}$, then

$$DNC(D(N), \partial N) = \partial N \times \mathbb{R} \times \{0\} \sqcup D(N) \times (0, 1]$$

- Consider the deformation of the inclusion $\{0\} \to \mathbb{R}$, i.e.,

$$DNC(\mathbb{R}, \{0\}) = \{0\} \times \mathbb{R} \times \{0\} \sqcup \mathbb{R} \times [0, 1]^* = \mathbb{R} \times [0, 1]$$

- Applying the deformation functor to the map of pair z, to get

$$DNC(z) = d_\nu z \sqcup (z \times Id) : \partial N \times \mathbb{R} \times \{0\} \sqcup D(N) \times (0, 1] \to \mathbb{R} \times [0, 1],$$

  where $d_\nu z : \partial N \times \mathbb{R} \to \mathbb{R} \times \{0\}$ is such that $d_\nu z(n, v) = v$. Therefore we get that in terms of the inverse image,

$$\begin{aligned}\overline{D} &= d_\nu z^{-1}([0, \infty) \times \{0\}) \cup (z \times Id)^{-1}([0, \infty) \times (0, 1]) \\ &= DNC(z)^{-1}([0, \infty) \times [0, 1]).\end{aligned}$$

We get the relation

$$\begin{aligned}\overline{D} &= (\partial N \times [0, 1)) \times \{0\} \sqcup N \times (0, 1] \\ &= \partial N \times \{0\} \times [0, 1] \sqcup \Big(\partial N \times (0, 1) \times \{0\} \sqcup N^0 \times (0, 1]\Big)\end{aligned}$$

Then we get the following commutative diagram

$$\begin{tikzcd}
K^*(\partial N) \arrow[r, "Th"] & K^{*+1}(\partial N \times (0,1)) \\
K^*(\partial N \times [0,1]) \arrow[u, two heads, "ev_0^h"] \arrow[d, two heads, "ev_1^h"'] & K^{*+1}(DNC(N^0, \partial N, l)) \arrow[u, "ev_0"'] \arrow[d, "ev_1"] \\
K^*(\partial N) \arrow[r, "\partial"'] & K^{*+1}(N^0)
\end{tikzcd}$$

where $ev_0^h \circ (ev_1^h)^{-1} = Id$ by homotopy invariance, showing that

$$\partial = ev_1 \circ ev_0^{-1} \circ Th = l!$$

Falta el argumento de pushforward para encajes como l en este caso.

5.3.7 Naturality with respect to closed subsets

Condiciones sobre la cohomologia Consider $f : M \to N$ a continuous map between manifolds without boundary. The pushforward $f! : \mathscr{H}_\Gamma^*(M) \to \mathscr{H}_\Gamma^{*-m+n}(N)$ is defined by the composition

$$\mathscr{H}_\Gamma^*(M) \xrightarrow{\tau_f} \mathscr{H}_\Gamma^{*+k}(T_f) \xrightarrow{e_0^{-1}} \mathscr{H}_\Gamma^{*+k}(D_f) \xrightarrow{e_1} \mathscr{H}_\Gamma^{*+k}(M^2 \times N) \xrightarrow{Mor} \mathscr{H}_\Gamma^{*+k}(N)$$

where

- The map $\tau_f : \mathscr{H}_\Gamma^*(M) \to \mathscr{H}_\Gamma^*(T_f)$ is the Thom isomorphism for the normal bundle $T_f := TM \oplus f^*TN \to M$.
- $D_f := DNC(M^2 \times N, M) = T_f \times \{0\} \sqcup ((M^2 \times N) \times (0,1])$ for the embedding $(\Delta, f) : M \to M^2 \times N$, with $\Delta : M \to M^2$ the diagonal map.
- The maps $e_i : \mathscr{H}_\Gamma^*(D_f) \to \mathscr{H}_\Gamma^*(C_i)$ for $i = 0, 1$, with $C_0 = T_f$ and $C_1 = M^2 \times N$, is defined by Como se definen estos mapas. We have that e_0 is an isomorphism.
- $Mor : \mathscr{H}_\Gamma^*(M^2 \times N) \to \mathscr{H}_\Gamma^*(N)$ is defined by La definicion de este mapa debe coincidir con la traza en K-teoria.

Now, we want to determine the compatibility of this definition with some maps related to restrictions to closed subspaces in the following way: if C_M and C_N are closed submanifolds of M and N respectively, such that the following diagram is commutative

$$\begin{tikzcd}
M \arrow[r, "f"] & N \\
C_M \arrow[u, "i_{C_M}"] \arrow[r, "f_C"'] & C_N \arrow[u, "i_{C_N}"']
\end{tikzcd}$$

therefore we get the diagram (Where $\triangle := * + k$)

$$\begin{tikzcd}
\mathscr{H}^*_\Gamma(M) \arrow[r, "\tau_f"] \arrow[d, "i^*_{C_M}"'] & \mathscr{H}^\triangle_\Gamma(T_f) \arrow[r, "e_0^{-1}"] \arrow[d, "A"'] & \mathscr{H}^\triangle_\Gamma(D_f) \arrow[r, "e_1"] \arrow[d, "D"] & \mathscr{H}^\triangle_\Gamma(M^2 \times N) \arrow[r, "Mor"] \arrow[d, "B"] & \mathscr{H}^\triangle_\Gamma(N) \arrow[d, "i^*_{C_N}"] \\
\mathscr{H}^*_\Gamma(C_M) \arrow[r, "\tau_{f_C}"'] & \mathscr{H}^\triangle_\Gamma(T_{f_C}) \arrow[r, "e_0^{-1}"'] & \mathscr{H}^\triangle_\Gamma(D_{f_C}) \arrow[r, "e_1"'] & \mathscr{H}^\triangle_\Gamma(C^2_M \times C_N) \arrow[r, "Mor"'] & \mathscr{H}^\triangle_\Gamma(C_N)
\end{tikzcd}$$

where the vertical maps are given as follows

- The map $D((i^2_{C_M}, i_{C_N}))^* : \mathscr{H}^*_\Gamma(D_f) \to \mathscr{H}^*_\Gamma(D_{f_C})$ is defined by the pullback of the map $D((i^2_{C_M}, i_{C_N})) : D_{f_C} \to D_f$, where the map $(i^2_{C_M}, i_{C_N})$ is constructed following the commutative diagram

$$\begin{tikzcd}
M \arrow[r, "(\Delta, f)"] & M^2 \times N \\
C_M \arrow[u, "i_{C_M}"] \arrow[r, "(\Delta_{C_M}, f_C)"'] & C^2_M \times C_N \arrow[u, "(i^2_{C_M}, i_{C_N})"']
\end{tikzcd}$$

  then,

  $D((i^2_{C_M}, i_{C_N})) : T_{f_C} \times \{0\} \sqcup ((C^2_M \times C_N) \times (0,1]) \to T_f \times \{0\} \sqcup ((M^2 \times N) \times (0,1])$

  Where

$$D((i^2_{C_M}, i_{C_N}))|_{T_{f_C}} = di_{C_M} \oplus f^* di_{C_N} = i_{TC_M} \oplus i_{f^*_C TC_N}$$

  and

$$D((i^2_{C_M}, i_{C_N}))|_{(C^2_M \times C_N) \times (0,1]} = (i^2_{C_M}, i_{C_N}) \times Id_{(0,1]}$$

  Debemos probar que $D((i^2_{C_M}, i_{C_N}))$ es un mapa propio.

- A Since $f : M \to N$ we get that $df : TM \to TN$ and $df_C : TC_M \to TC_N$, therefore we get the map $df \oplus df_C : T_f \to T_{f_C}$.

About the commutativity of the squares of the previous diagram (from left to right)

1. Naturality of the Thom isomorphism.
2. Restriction of the deformation to the normal cone to its zero component.
3. Restriction of the deformation to the normal cone to its one component.
4. Naturality of Mor (Debemos revisar que se tenga la naturalidad en la traza en K-teoria, donde naturalidad significa la invarianza entre el orden en que se aplica traza o aplicar restriccion.)

Therefore we can conclude that in this case $i^*_{C_N} \circ f! = f_C! \circ i^*_{C_M}$.

5.4 Review of pushforward structure in Equivariant K-theory

- The groupoids that are involved are the same but with the action of Γ in such a way that for instance in the first step $K^*_\Gamma(M) \cong K^*((\Gamma \ltimes M) \rightrightarrows M)$ (wich is an isomorphism for proper actions requiere cita), the groupoid is the $(\Gamma \ltimes M) \rightrightarrows M$ has the structure $s((r, m)) = m$ and $t((r, m)) = r \cdot m$.
- Same use os maps as in the non equivariant of the previos subsection.
- The morita equivalence are preserved by the applying action of Γ.
- We have to describe the ways and modifications that we have to prove in order to preserve the propositions with the action.

5.5 Review Baum-Douglas Homology

5.5.1 Analytical K-holomogy

Seccion 4....

Definition 5.9. *$K^a(X)$ is defined as the equivalence clases of 5-tuples $(\mathcal{H}_0, \psi_0, \mathcal{H}_1, \psi_1, T)$, under the relation $\sim$, and where*

- *$\mathcal{H}_i$ is a separable Hilbert space for $i = 0, 1$.*
- *$\psi_i : C(X) \to L(\mathcal{H}_i)$ is a unital algebra $*-$homomorphism.*
- *$T : \mathcal{H}_0 \to \mathcal{H}_1$ is a bounded Fredholm operator with $T \circ \psi_0(f) - \psi_1(f) \circ T$ being a compact operator for any $f \in C(X)$*

$$\begin{tikzcd}
\mathcal{H}_0 \arrow[r, "T"] \arrow[dr] & \mathcal{H}_1 \arrow[d, "\psi_1(f)"] \\
\mathcal{H}_0 \arrow[u, "\psi_0(f)"] \arrow[ur] & \mathcal{H}_1
\end{tikzcd}$$

The relation $\sim$ is generated such that

- *Isomorphic 5-tuples are equivalent, where for a pair of cycles $(\mathcal{H}_0, \psi_0, \mathcal{H}_1, \psi_1, T)$ and $(\mathcal{H}'_0, \psi'_0, \mathcal{H}'_1, \psi'_1, T')$, an isomorphism is given by operators $U_i; \mathcal{H}_i \to \mathcal{H}'_i$, for $i = 0, 1$ such that*
  - *$T' = U_1 T U_0^*$.*
  - *$\psi_i(f) = U_i \psi_i(f) U_i^*$.*
- *Direct sum with a trivial object:*

5.5.2 Spin^C-structures

The group Spin(n) is the double cover of SO(n), with projection

$$\eta : \mathrm{Spin}(n) \cong \mathbb{Z}_2 \times \mathrm{SO}(n) \to \mathrm{SO}(n).$$

Denote $\epsilon \in Ker(\eta) \cong \mathbb{Z}_2$ the non trivial element and consider

$$\mathrm{Spin}^{\mathrm{c}}(n) = (\mathrm{Spin}(n) \times S^1)/\Gamma,$$

where $\Gamma = \{(\epsilon, -1), (1, 1)\}$ acts on Spin$(n) \times S^1$ by aclarar accion. For a vector bundle $E \to Y$, consider the fiber bundle $F(E) \to Y$ whose fibre at $p \in Y$ is the set of all vector space bases of E_p (considering Y connected and $\dim_{\mathbb{R}}(E_p) \leq 3$). When $F(E)$ is connected, we say that the vector bundle is orientable, but when $F(E)$ has exactly 2 connected components, we say that the vector bundle $E \to Y$ is orientable, where a choice of one of these two connected components is an orientation for the vector bundle, denoted $F^+(E)$ (where there is an homeomorphism $F^+(E)_p \cong GL^+(n, \mathbb{R})$). Since the cohomology groups of $GL^+(n, \mathbb{R})$ are

$$H^0(GL^+(n, \mathbb{R})) = \mathbb{Z}, \quad H^1(GL^+(n, \mathbb{R})) = 0, \quad H^2(GL^+(n, \mathbb{R})) = \mathbb{Z}_2,$$

then for the inclusion $i_{F^+(E)_p} : F^+(E)_p \to F^+(E)$ we get the cohomology induced map

$$i^*_{F^+(E)_p} : H^2(F^+(E)) \to \mathbb{Z}_2 = \{1, -1\}.$$

Definition 5.10. *A* Spin^c*-structure for a vector bundle $E \to Y$ is*

- *An orientation for E.*
- *A cohomology class $u \in (i^*_{F^+(E)_p})^{-1}(-1)$ for any $p \in Y$.*

We say that a smooth manifold M is a Spin^c manifold if there is a Spin^c-structure for the tangent bundle $TM \to M$. If M is a Spin^c manifold with $\partial M \neq \emptyset$, then ∂M is a Spin^c manifold too.

5.5.3 Clutching construction

Let M be a Spin^c manifold, and $H \to M$ be a Spin^c vector bundle with even dimensional fibres. Considering a positive definite symmetric inner product for $H \oplus (M \times \mathbb{R}) \to M$, define

$$\hat{M} := S(H \oplus (M \times \mathbb{R})),$$

which is a even dimensional (A mi me da impar) sphere bundle $\rho : \hat{M} \to M$ and it is a Spin^c manifold (Como se obtiene esta estructura spinc). From [**?**] since $n = \dim_{\mathbb{R}}(H_p)$ is even, Spin$^c(n)$ has two 1/2-Spin representations (no es $n/2$?) complex vector bundles $H_\pm \to M$. A pairing

$$H \times H_+ \to H_-$$

is given by Clifford multiplication, which defines a vector bundle map

$$\sigma : H_0 := \rho^*(H_+) \to H_1 := \rho^*(H_-),$$

where σ is an isomorphism off the zero section of H(No se que significa eso). Then we can identify

$$\hat{M} = B(H) \bigcup_{S(H)} B(H),$$

where $B(H)$ and $S(H)$ represent unit ball bundle and sphere bundle of H, respectively. With these we construct $(\hat{M}, \hat{H}, \rho)$, where we define

$$\hat{H} = H_0 \cup_\sigma H_1 \to \hat{M}.$$

For each $p \in M$, the restriction of $\hat{H}$ to the even-dimensional sphere $\rho^{-1}(p)$, is the Bott generator vector bundle on $\rho^{-1}(p)$.

5.5.4 Geometrical K-homology

Definition 5.11. *A K-cycle on a topological space X is a triple (M, E, ϕ) such that:*

1. *M is a compact Spin^C-manifold without boundary.*
2. *$E \to M$ is a complex vector bundle on M.*
3. *$\phi : M \to X$ is a continuous map.*

Two K-cycles are considered isomorphic if there is a diffeomorphism $h : M \to M'$ such that h preserves the Spin^c-structures, $h^*(E') \cong E$, and $\phi = \phi' \circ h$. The collection of K-cycles is denoted by $\Gamma(X)$.

Definition 5.12. *Define the geometric K-homology of X by $K^{geo}(X) = \Gamma(X)/\sim$, where the relation $\sim$ is an equivalence relation generated by bordism, direct sum and vector bundle modification.*

The vector bundle modification of (M, E, ϕ) by a given smooth Spin^c vector bundle $H \to M$ with even dimensional fibres, is the cycle $(\hat{M}, \hat{H} \otimes \rho^*(E), \phi \circ \rho)$, where

$$\begin{array}{ccccc} \hat{H} \otimes \rho^*(E) & & E & & \\ \downarrow & & \downarrow & & \\ \hat{M} & \xrightarrow{\rho} & M & \xrightarrow{\phi} & X \end{array}$$

$\hat{H}$ is the clutching construction constructed in the previous section.

About $K^{geo}(X)$ we get

1. It is an abelian group with operation given by disjoint union.

2. It define a covariant functor.

3. $K^{geo}(X) = K_0^{geo}(X) \oplus K_1^{geo}(X)$, where $K_0^{geo}(X)$ and $K_1^{geo}(X)$ are the subgroup of cycles (M, E, ϕ) with M even and odd dimensional, respectively.

4. There is a bilinear pairing $K^0(X) \otimes K^{geo}(X) \to K^{geo}(X)$ given by teh cup product $F \cap (M, E, \phi) = (M, E \otimes \phi^* F, \phi)$.

   Example 5.13. *$K_0^{geo}(\{*\}) = \mathbb{Z}$ and $K_1^{geo}(\{*\}) = 0$, by Bott periodicity.*

5.5.5 The isomorphism between Analytical and geometrical K - homologies

For a topological space X, in [**?**] it is defined an isomorphism $\mu : K^g(X) \to K^a(X)$ by

$$\mu(M, E, \phi) = \phi_*(E \cap [D]),$$

where:

1. D is the Dirac operator constructed in the following way: Denote the complexified cotangent bundle by $T^* = \mathbb{C} \otimes_{\mathbb{R}} TM^*$. Denote by $C^\infty(V)$ the vector space of all smooth sections of a smooth vector bundle $V \to M$. Definition of the Dirac operator depends on the parity of $n = dim(M)$:

   - If n is even, considering $F_\pm \to M$ the two $\mathrm{Spin}^c(n)$ smooth complex vector bundles given by the Spin^c structure on $TM \to M$Me encantaria poder explicar la manera en que la estructura spin aparece en las estructuras spinc. , the Dirac operator $D : C^\infty(F_+) \to C^\infty(F_-)$ is defined as the composition $D := \gamma_* \circ \nabla$, where
     - $\nabla : C^\infty(F_+) \to C^\infty(T^* \otimes F_+)$ is a connection[5] for $F_+ \to M$.
     - $\gamma : T^* \otimes F_+ \to F_-$ is given by Clifford multiplication, getting $\gamma_*(s) = \gamma \circ s : M \to F_-$ for any $s \in C^\infty(T^* \otimes F_+)$.
   - If n is odd, the Spin^c structure on the tangent bundle of M associates to the Spin representation of $\mathrm{Spin}^c(n)$ a smoth complex vector bundle $F \to M$ and in the same fashion of even case, the Dirac operator is defined as the composition:

   $$D = \gamma_* \circ \nabla : C^\infty(F) \to C^\infty(T^* \otimes F) \to C^\infty(F).$$

   In both cases and for any $E \to M$ smooth vector bundle, choosing a connection for $F \otimes E \to M$, the operator $D \otimes I_E : C^\infty(\widetilde{F} \otimes E) \to C^\infty(\overline{F} \otimes E)$ is the composition

   $$D \otimes I_E = \gamma_* \circ \nabla : C^\infty(\widetilde{F} \otimes E) \to C^\infty(T^* \otimes \overline{F} \otimes E) \to C^\infty(\overline{F} \otimes E)$$

   where $\widetilde{F} = F_+$ and $\overline{F} = F_-$ in even case, and $\tilde{F} = \overline{F} = F$ in the odd case.

[5] A $\mathbb{C}$-linear map such that $\nabla(f \cdot s) = df \otimes s + f \cdot \nabla(s)$ for $f \in C^\infty(M)$, $s \in C^\infty(F_+)$, where $d : C^\infty(\wedge^j T^*) \to (\wedge^{j+1} T^*)$, is the de Rham operator.

2. Following Section 5 in [BD82], for a Dirac Operator[6] $[D] \in K^a(M)$ is determinded as

$$[D] := (L^2(E_0), \psi_0, L^2(E_1), \psi_1, V),$$

where for $i = 0, 1$:

   - For the vector bundles $E_i \to X$, then $L^2(E_i)$ is the completion of $C^\infty(E_i)$ respect to the inner product defined from a choosed measure μ on M, and Hermitian structures on E_i as follows:

$$\langle u, v\rangle = \int_M \langle u(p), v(p)\rangle \, d\mu, \qquad u, v \in C^\infty(E_i).$$

   - $V : L^2(E_0) \to L^2(E_1)$ is the partial isometry part in the polar decomposition of the closable operator $D : L^2(E_0) \to L^2(E_1)$ (Buscar referencia para garantizar su existencia y unicidad), being this a bounded operator.
   - The maps ψ_i are constructed in the following way

$$C(X) \xrightarrow{\quad\quad\psi_i\quad\quad} \mathrm{L}(L^2(E_i))$$

$$f \longmapsto \psi_i(f) : L^2(E_i) \to L^2(E_i) : u \mapsto [\psi_i(f)](u)$$

   (Seria chevere ver porque esta es cuadrado integrable) such that for any $e \in E_i$, then

$$[\psi_i(f)](u) : E_i \to E_i : e \mapsto ([\psi_i(f)](u))(e) := f(\rho_{E_i}(e)) \cdot u(e)$$

3. The cap product is defined by ... (Seccion 5, y los profes recomiendan leer el producto de kasparov)

4. $\phi_* : K^a(M) \to K^a(X)$ is the induced map by $\phi : M \to X$.

5.6 Equivariant K-Homology

Definition 5.14 ([BHS07], p. 12)**.** *A complex spinor bundle for V, where $V \to M$ being an euclidean vector bundle over M of rank p, is a p-multigraded Dirac bundle S_V which is locally isomorphic to the trivial bundle with fiber the complex Clifford algebra $\mathbb{C}_p$, where:*

1. *A p-multigraded Dirac bundle structure on V* (see [BHS07], p. 6) *is a smooth $\mathbb{Z}_2$-graded, Hermitian vector bundle S over M together with the following data:*

[6] In [BD82], the formulation is in general for any elliptic pseudo differential operator.

(a) *An $\mathbb{R}$-linear morphism of vector bundles $V \to End(S)$ which associates to each vector $v \in V_x$ a skew-adjoint, odd-graded endomorphism $u \mapsto v \cdot u := v(u)$ of S_x, in such a way that*

$$v \cdot v \cdot u = -||v||^2 u.$$

(b) *A family of skew-adjoint, odd-graded endomorphism $\varepsilon_1, ..., \varepsilon_p$ of S such that*

$$\varepsilon_j = -\varepsilon_j^*, \quad \varepsilon_j^2 = -1, \quad \varepsilon_i\varepsilon_j + \varepsilon_j\varepsilon_i = 0 (i \neq j),$$

and such that each ε_j commutes with each operator $u \mapsto v \cdot u$.

If M is a smooth manifold (possibly with boundary) then by a spin^c structure on M we mean a pair consisting of a Riemannian metric on M and a complex spinor bundle S_M for TM. Then for each $x \in M$, there exist an open $x \in U_x \subseteq M$ with the following diagram

$$\begin{tikzcd}
S_M|_{U_x} \cong U_x \times \mathbb{C}_p \arrow[d] & S_M \arrow[dr] & TM \arrow[r] \arrow[d] & End(S_M) \arrow[dl] \\
U_x \arrow[rr, hook] & & M &
\end{tikzcd}$$

*A Γ-spin^c manifold is a manifold with a spin^c-structure given in terms of a complex spinor bundle S_M for TM with a Γ-action listed to and compatible with all the structure.*Ser más específico

Definition 5.15 ([BOOSW10], p. 5)**.** *The equivariant K-homology respect to a discrete group Γ*[7] *of a Γ-CW pair (X, Y), denoted $K_*^{\Gamma, geom}(X, Y)$ is the set of isomorphism classes of* **cycles** *identified by a equivalence* **relation** $\sim$*, those defined as follows:*

1. *A* **cycle** *is a triple (M, E, f), where*
   - *M is a compact smooth Γ-spin^c manifold.*
   - *E is a Γ-equivariant Hermitian vector bundle on M.*
   - *$f : M \to X$ is a continuous Γ-equivariant map such that $f(\partial M) \subset Y$.*
2. *An isomorphism between two cycles (M, E, f) and (M', E', f') is a smooth Γ-map $\phi : M \to M'$ such that*

   (a) *push or pull back spin^c structure.*

   (b) *$\phi^*(E') = E$.*

   (c) *$f = f' \circ \phi$.*

[7] This is defined in general for a compact Lie group G, see [BOOSW10].

3. The equivalence **relation** *generated by the following conditions:*

- $(M, E, f) \sqcup (M, E', f) \sim (M, E \oplus E', f)$.

$$\begin{tikzcd} E \arrow[r, hook] \arrow[dr] & E \oplus E' \arrow[d] & E' \arrow[l, hook'] \arrow[dl] \\ & M \arrow[r, "f"] & X \end{tikzcd}$$

- $(M, E, f) \sim (M', E', f')$ *if there is a bordism between* (M, E, f) *and* $-(M', E', f')$[8], *where a bordism beetwen* K*-cycles over* (X, Y)*, identified as* $M_+ = g^{-1}[1, \infty)$ *and* $M_+ = g^{-1}(-\infty, -1]$*, is the following bunch of data:*
  - *A smooth, compact* Γ*-manifold* L*, equipped with a* Γ*-spin*^{c}*-structure.*
  - *A smooth, Hermitian* Γ*-vector bundle* F *over* L*.*
  - *A continuous* Γ*-map* $\Phi : L \to X$*.*
  - *A smooth* Γ*-invariant map* $g : \partial L \to \mathbb{R}$ *for which* ± 1 *are regular values, and for wich* $\Phi[g^{-1}[-1, 1]] \subseteq Y$*.*

$$\begin{tikzcd} \mathbb{R} & & F \arrow[d] & \\ \partial L \arrow[u, "g"] \arrow[rr, hook] & & L \arrow[r, "\Phi"] & X \\ g^{-1}([-1,1]) \arrow[u, hook] \arrow[urr, hook, "i"] \arrow[rrr, "\Phi \circ i"'] & & & Y \arrow[u, hook] \end{tikzcd}$$

then we mean that between (M, E, f) *and* $-(M', E', f')$ *there exist a bordism such that* $g^{-1}(\{1\}) = M$*,* $g^{-1}(\{-1\}) = -M'$*,* $i^* F \cong E \sqcup E' \to g^{-1}(\{-1, 1\}) = M \sqcup (-M')$ *Revisar*

- $(M, E, f) \sim (Z, F \otimes \pi^* E, f \circ \pi)$*, as in the following diagram*

$$\begin{tikzcd} F \otimes \pi^* E \arrow[d] & E \arrow[d] & \\ Z \arrow[r, "\pi"] & M \arrow[r, "f"] & X \end{tikzcd}$$

being the later the modification of a Γ*-spin*^{c} *bundle* $W \to M$*, where:*

(a) Z *is the unit sphere bundle of the bundle* $\pi : Z \subset 1 \oplus W \to M$*.*

$$\begin{tikzcd} Z \arrow[r, hook] \arrow[dr, "\pi"'] & 1 \oplus W \arrow[d] \\ & M \end{tikzcd}$$

[8]By $-(M', E', f')$ we mean the K-cycle $(-M', E', f')$ where $-M'$ is the manifold M' equipped with the opposite spin^c structure of M'.

(b) F is obtained as $\pi^* S^*_{W,+}$ over the northern hemisphere[9] *D^+ of Z and $\pi^* S^*_{W,+}$ over the southern hemisphere D^- of Z by gluing along the intersection, the unit sphere bundle of W, using Clifford multiplication with the respective vector of W.*

$$
\begin{tikzcd}
\pi^* S^*_{W,\pm} \arrow[d] & S^*_{W,\pm} \arrow[d] & \Rightarrow & F \cong \pi^* S^*_{W,+} \sqcup_{t=0} \pi^* S^*_{W,-} \arrow[d] \\
D^{\pm} \arrow[r, "\pi"'] & M & & Z \cong D^+ \sqcup_{t=0} D^-
\end{tikzcd}
$$

5.7 Periodic Cyclic Homology

5.8 Pushforward homology as a quotient group

Consider the free abelian group generated by the collection of isomorphic cycles $(M, \partial M, \pi_M, x_M)$ related to (X, A) just like in Definition **??**, denoted by $\mathbb{Z}_{\mathscr{H}^*_\Gamma}[Cyc]$, and consider the subgroup $\mathcal{R}_{\mathscr{H}^*_\Gamma}[Cyc]$ generated by:

- $[M \sqcup N, \partial M \sqcup \partial N, \pi_M \sqcup \pi_N, x_M \oplus x_N] - [M, \partial M, \pi_M, x_M] - [N, \partial N, \pi_N, x_N]$.
- $[M, \partial M, \pi_N \circ f, x_M] - [N, \partial N, \pi_N, f!(x_M)]$, where $f : (M, \partial M) \to (N, \partial N)$ is a proper Γ-equivariant, orientation preserving smooth map.
- $[M, \partial M, \pi_M, x_M + x'_M] - [M, \partial M, \pi_M, x_M] - [M, \partial M, \pi_M, x'_M]$.

6 Assembly map

In [RWW23] there exist a way to relate cycles using a well defined group in the case of the pair $(\{*\}, \{*\})$ (wich is not a Γ proper pair).

7 Generalized $\mathscr{H}^*_\Gamma$-oriented bundles

For a fixed $\mathscr{H}^*_\Gamma$-cohomology, we will consider the $\mathscr{H}^*_\Gamma$-orientation of Γ-vector bundles as a class of Γ-vector bundles such that the class its closed by

1. smooth orientation inverse (classical orientation in the de Rham cohomology);
2. Witney sum;
3. Pull backs;

such that for each of the vector bundle $E \to M$ of this class there exist an isomorphism $\tau_{E,M} : \mathscr{H}^*_\Gamma(M) \to \mathscr{H}^*_\Gamma(E)$ such that

1. For $E = M \times 0$, $\tau_{E,M} = Id_{\mathscr{H}^*(M)}$.

[9] The notion of Hemisphere has sense identifying Z as the pairs $(t, w) \in 1 \oplus W$ where $t \in [-1, 1]$ and $t^2 + |w|^2 = 1$, and the hemispheres are defined by the sign of t.

2. For $f : N \to M$, and an $\mathscr{H}$-vector bundle $E \to N$, $f! \circ \tau_{f^*E,M} = \tau_{E,N} \circ f!$.

3. For a pair of vector bundles $E \to M$ and $F \to M$, and a vector bundle V over E and F at the same time, such that $\rho_E \circ \rho_{V,E} = \rho_F \circ \rho_{V,F}$, then $\tau_{V,E} \circ \tau_{E,M} = \tau_{V,F} \circ \tau_{F,M}$.

Proposition 7.1. *If $E = -F$ (opposite in the sense of de Rham cohomology), then $\tau_{E,M} = -\tau_{F,M}$.*

8 Examples

- $\Gamma = \{e\}$.
- $\mathscr{H}_*^{pf}(M, \partial M; \Gamma) \cong \mathscr{H}_\Gamma^*(M, \partial M)$.

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